RUTGERS

Rutgers University NHETC

Physics-Informed Neural Networks for Galactic Dark Matter

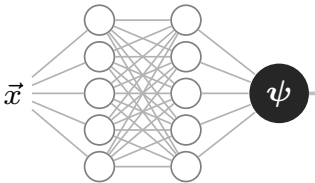
PINNs for Gaia kinematics and the diffuse γ-ray background

Eric Putney
Rutgers University, NHETC

Westlake Seminar

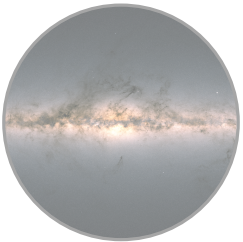
12 October 2026

Roadmap



Physics-informed neural networks

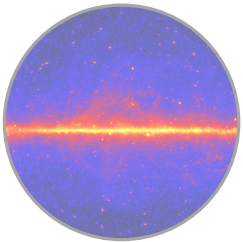
The PDE residual as the loss function



Part I: ClearPotential

Dark matter from the kinematics of 5.8 M *Gaia* stars

with M. Buckley, S. H. Lim, D. Shih



Part II: PINNCR

Diffuse γ-ray background to the GeV excess

with Y. Park, D. Shih, M. Buckley, T. Slatyer, C. Chen

Images: ESA/Gaia/DPAC; NASA/DOE/Fermi-LAT Collaboration

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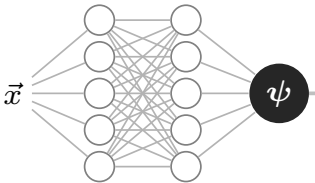

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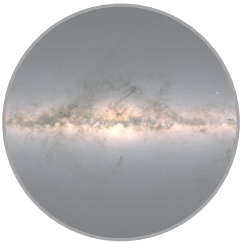

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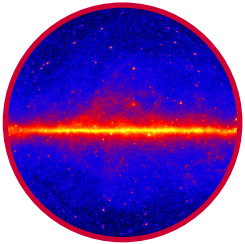

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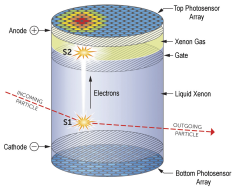
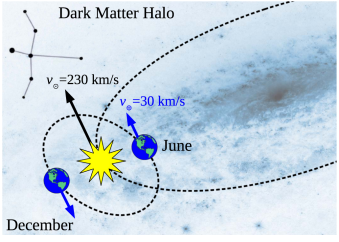
The Milky Way as a dark matter laboratory

If DM couples to the SM, several channels:

- **Direct** detection (XENONnT, LZ): rate $\propto$ the local DM density $\rho_{\mathrm{DM},\odot}$





A xenon time-projection chamber, and the halo wind it looks for.

LZ Collaboration (Aalbers+22); Schumann 2019

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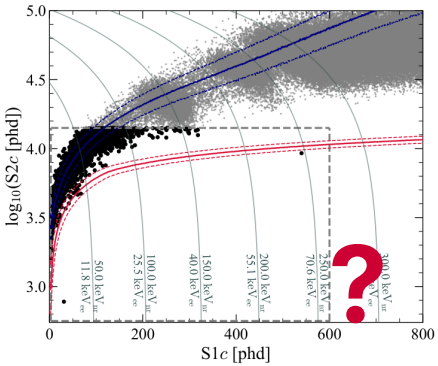

LZ science sample; the one event sits in the dashed box.

LZ Collaboration, arXiv:2609.02823, Fig. 4

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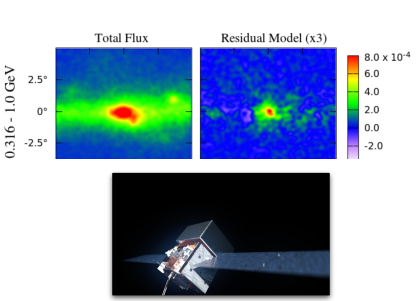
- **Indirect** detection (Fermi-LAT): the Galactic Center GeV excess, over a large astrophysical background



The GeV excess and a DM-annihilation template; Fermi-LAT.

Daylan et al. 2014; NASA

The Milky Way as a dark matter laboratory

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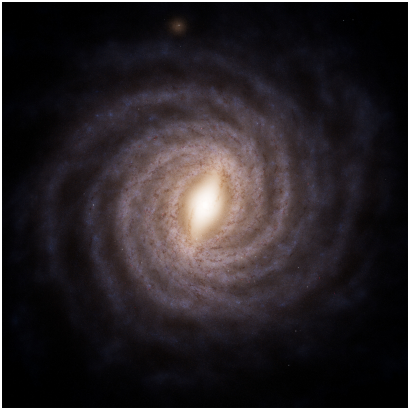
Either way, the constraints depend on the Galactic distribution of DM, known only through gravity.

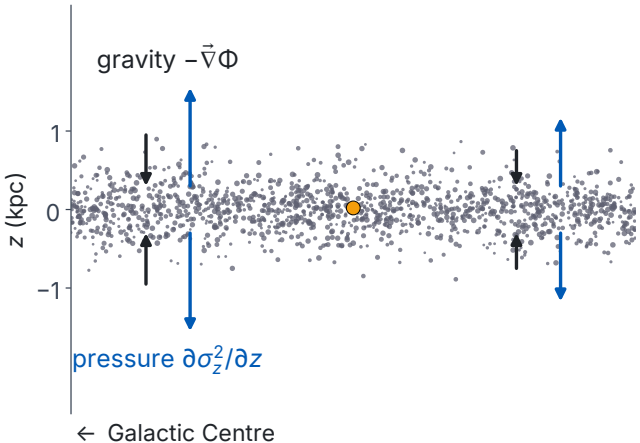


ESA/Gaia/DPAC, S. Payne-Wardenaar

The Oort problem

Classic dynamics problem: Oort estimated the local DM density in *1932*

- Stellar disk ≈ equilibrated gas
- Pressure ($\partial\sigma_z^2/\partial z$) pushes out, gravity ($-\vec{\nabla}\Phi$) pulls in



Vertical balance: the measured pressure out, gravity in; with DM, more pull to match it. Sun in gold.

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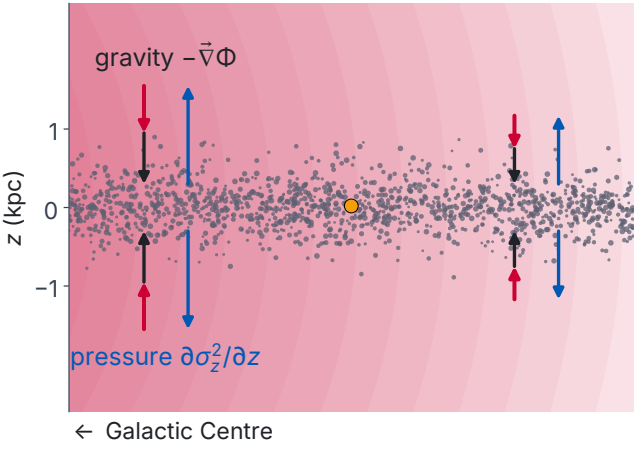
Classic dynamics problem: Oort estimated the local DM density in *1932*

- Stellar disk $\approx$ equilibrated gas
- Pressure ($\partial\sigma_z^2/\partial z$) pushes out, gravity ($-\vec{\nabla}\Phi$) pulls in
- Stars trace the gravitational potential of **all** mass: stars, gas, DM

Vertical Jeans equation:

- moments of f, assumes symmetries
- not the general EoM



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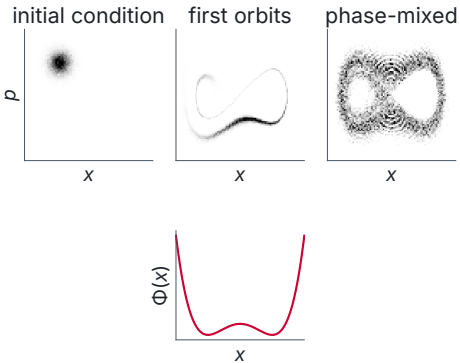
The collisionless Boltzmann equation

Stars as a collisionless fluid in 6D phase space, $f(\vec{x}, \vec{v})$:

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \vec{\nabla}_x f - \vec{\nabla}_x \Phi \cdot \vec{\nabla}_v f = 0$$

- f fixes Φ; Poisson then gives ρ



A blob of stars in a toy double well phase-mixes into an equilibrium f.

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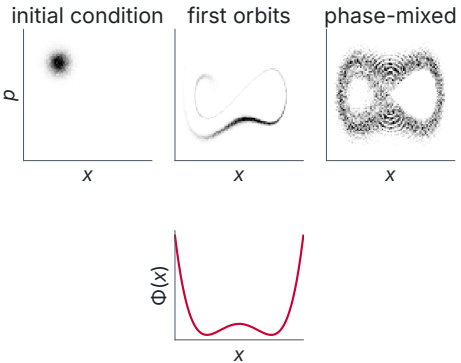
- Not perfectly relaxed (LMC, bar, spirals): assume it, test afterwards (backup)



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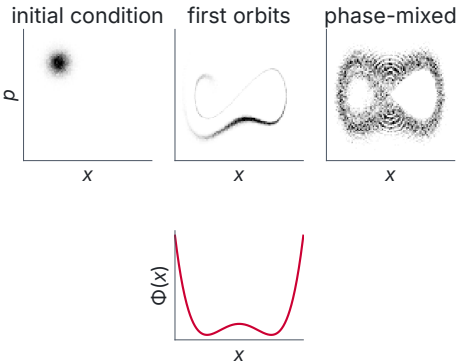
- Equilibrium ($\partial_t f = 0$): orbits ~ 100 Myr $\ll$ 10 Gyr; one snapshot suffices
- Not perfectly relaxed (LMC, bar, spirals): assume it, test afterwards (backup)
- Needs f and its derivatives in 6D: millions of stars, a flexible model of f

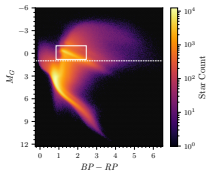


A blob of stars in a toy double well phase-mixes into an equilibrium f.

Measuring f: *Gaia* DR3

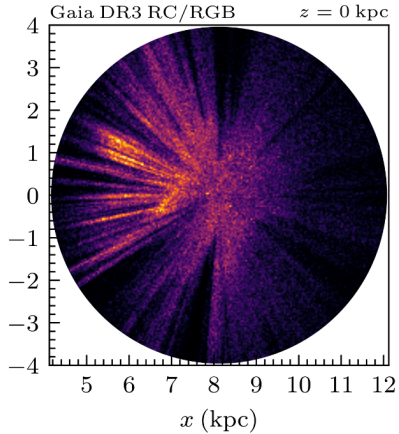
Gaia (2014–2025) mapped ∼2 B stars; DR3 (2022) uses the first 34 months:

- 33.8 M stars with full 6D phase space (1.46 B with 5D astrometry)
- Red clump + RGB tracers: 5.8 M within 4 kpc



$G_{\rm rvs}$ magnitude cut.

Lim et al. 2023



Tracers in the Galactic plane, Sun at center.

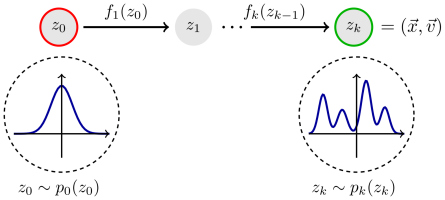
Putney et al. 2024

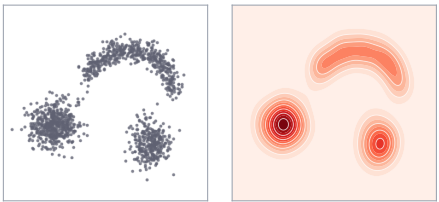
Normalizing flows

Normalizing flow: generative density estimation:

$$\vec{h} \to \vec{u} = g(\vec{h}), \quad p(\vec{u}) = p(g^{-1}(\vec{u})) \left| \det \frac{dg}{d\vec{h}} \right|^{-1}$$

- Invertible map g, latent $\vec{h} \to$ data $\vec{u}$, trained by maximum likelihood
- Smooth, differentiable $p(\vec{u})$: $\vec{\nabla}_x \ln f$, $\vec{\nabla}_v \ln f$ by autograd

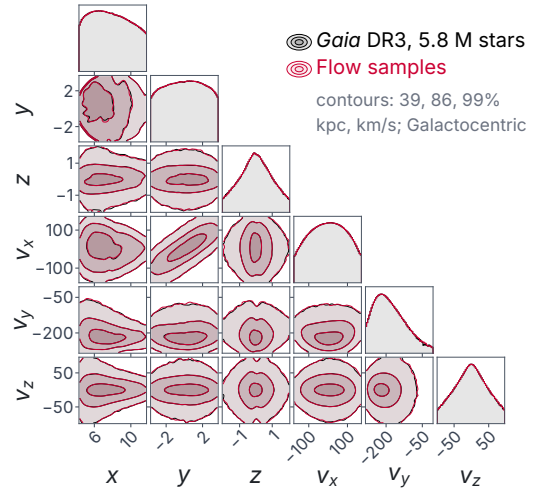




A 2D toy: points $\{\vec{u}_i\}$ and the flow's $p(\vec{u})$.

The flow fit to *Gaia* DR3

- $f = n(\vec{x})\, p(\vec{v}\,|\,\vec{x})$: two masked autoregressive flows, exact likelihood (backup)
- Reproduces the 6D marginals of 5.8 M stars



Gaia DR3 (black) vs. flow samples (red). *Lim et al. 2023*

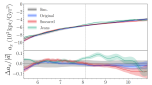
Solving the CBE for Φ

$\vec{a}(\vec{x})$ is independent of $\vec{v}$, and the flow gives any number of $\vec{v}$ at each $\vec{x}$:

$$\vec{v}_i \cdot \vec{\nabla}_x \ln f(\vec{x}, \vec{v}_i) - \vec{\nabla}_x \Phi \cdot \vec{\nabla}_v \ln f(\vec{x}, \vec{v}_i) = 0, \qquad i = 1, 2, \ldots$$

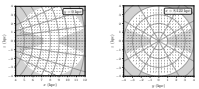
- Pointwise: least squares for $\vec{a}(\vec{x})$ at each $\vec{x}$ (N velocities, 3 unknowns) [Buckley+22, Lim+23]
- But Φ should be smooth and $\vec{a}$ curl-free: pointwise is expensive, incorrect

h277 hydro simulation: $\vec{a}$ to 1–5%



Buckley et al. 2022, arXiv:2205.01129

First application to data: *Gaia* DR3, 4 kpc



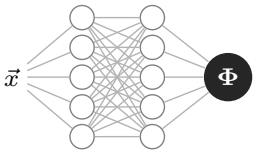
Lim et al. 2023, arXiv:2305.13358

Solving the CBE for Φ with a PINN

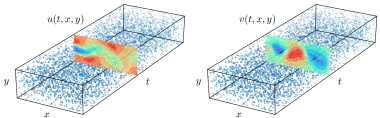
$\vec{a}(\vec{x})$ is independent of $\vec{v}$, and the flow gives any number of $\vec{v}$ at each $\vec{x}$:

$$\vec{v}_i \cdot \vec{\nabla}_x \ln f(\vec{x}, \vec{v}_i) - \vec{\nabla}_x \Phi \cdot \vec{\nabla}_v \ln f(\vec{x}, \vec{v}_i) = 0, \qquad i = 1, 2, \ldots$$

- Pointwise: least squares for $\vec{a}(\vec{x})$ at each $\vec{x}$ (N velocities, 3 unknowns) [Buckley+22, Lim+23]
- But Φ should be smooth and $\vec{a}$ curl-free: pointwise is expensive, incorrect
- PINN: $\Phi_\vartheta(\vec{x})$ a neural network, the CBE residual is the loss, $\mathcal{L}_\vartheta = \left\langle |\mathrm{CBE}|^2 \right\rangle_{\vec{x},\vec{v}}$
- No grid: random points in the volume, held-out points to validate
- $\vec{a} = -\vec{\nabla}\Phi_\vartheta$ curl-free by construction



$\Phi_\vartheta(\vec{x})$ as a network; CBE residual is the loss.



Navier–Stokes. *Raissi et al., arXiv:1711.10566*

Learning Φ from a snapshot of phase space

Pairing a flow for f with a network for Φ on the CBE goes back to 2020:

2020 — **Deep Potential** Green & Ting: flow for f, network for Φ, CBE residual as loss

2021–22 — **Charting accelerations** An, Naik, Evans & Burrage: when Φ is unique

2022–23 — **Pointwise CBE** Buckley+22 on a simulation; Lim+23: first application to real data, *Gaia* DR3

2023–25 — **Deep Potential, *Gaia*** Green, Ting & Kamdar 23; Kalda & Green 25, with the pattern speed

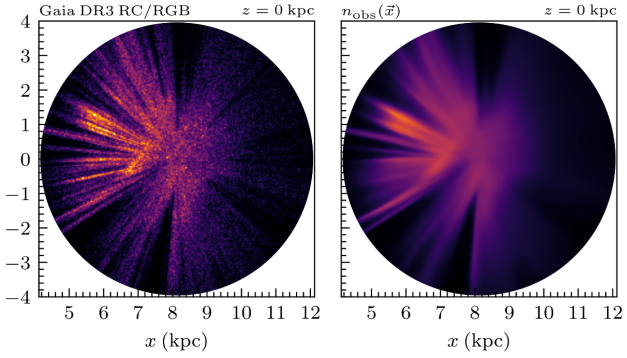
2024–25 — ClearPotential Φ-network again, plus dust (Putney+24, 25)

New here: dust ϵ learned inside the CBE · a 4 kpc volume · a full error covariance.

Dust removes stars

Optical surveys are obscured by dust lanes:

- Dims stars below *Gaia*'s radial-velocity limit ($G_{\rm RVS} \lesssim 14$)
- Missing stars bias f: a dust map cannot put them back



Gaia stars (left) vs. the flow's $n_{\rm obs}$ (right): the dust shadows are in f.

Putney et al. 2024

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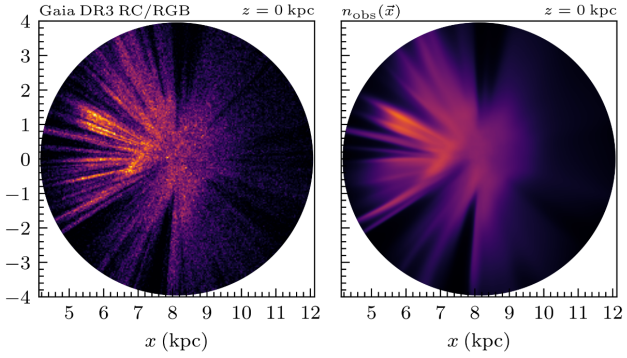
- Missing stars bias f: a dust map cannot put them back
- Worst in the midplane: derivatives of f amplify the shadows, biasing $\vec{a}$ and ρ



Gaia stars (left) vs. the flow's $n_{\rm obs}$ (right): the dust shadows are in f.

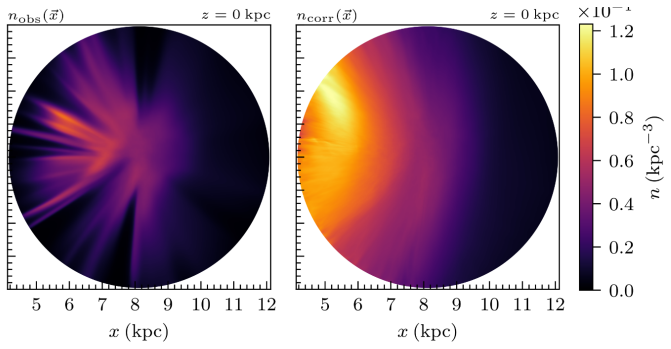
Putney et al. 2024

The dust-efficiency formalism

Relate the observed $f_{\rm obs}$ to the true f by a position-dependent efficiency $\epsilon(\vec{x})$:

$$f_{\rm obs}(\vec{x},\vec{v}) = \epsilon(\vec{x})\, f(\vec{x},\vec{v}) \qquad \Longrightarrow \qquad \vec{v}\cdot\vec{\nabla}\ln f_{\rm obs} - \vec{v}\cdot\vec{\nabla}\ln\epsilon - \vec{\nabla}\Phi\cdot\vec{\nabla}_v \ln f_{\rm obs} = 0$$

- **Second network** $\ln \epsilon_{\boldsymbol{\theta}}(\vec{x})$, same CBE loss



The flow's $n_{\rm obs}$ (left) and the dust-corrected $n_{\rm obs}/\epsilon$ (right) at $z = 0$.

Putney et al. 2024

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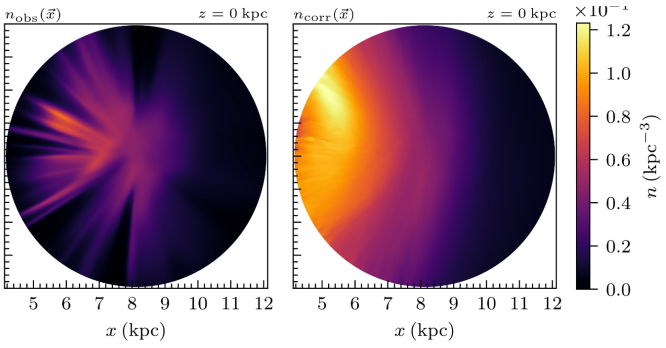
- **Second network** $\ln \epsilon_{\boldsymbol{\theta}}(\vec{x})$, same CBE loss
- **Velocity-independent**: dust removes stars by position
- **Degeneracy broken**: $\vec{\nabla}\ln\epsilon$ couples to $\vec{v}$, $\vec{\nabla}\Phi$ to $\vec{\nabla}_v \ln f_{\rm obs}$



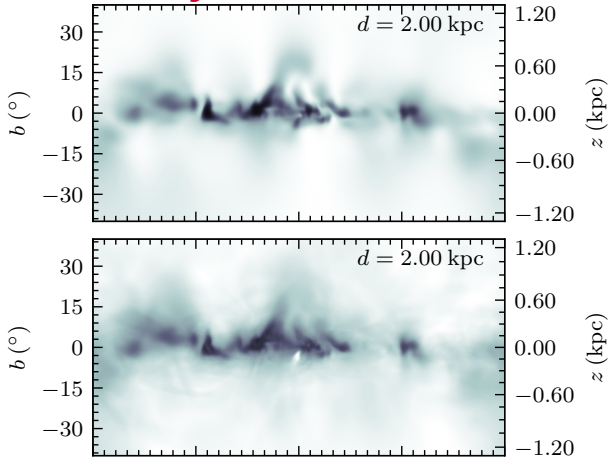
The flow's $n_{\rm obs}$ (left) and the dust-corrected $n_{\rm obs}/\epsilon$ (right) at $z = 0$.

Putney et al. 2024

Validating the learned dust efficiency

Independent check: $\epsilon_{\rm L22}$ from the Lallement+22 3D dust map

- Red clump returns above the completeness cut



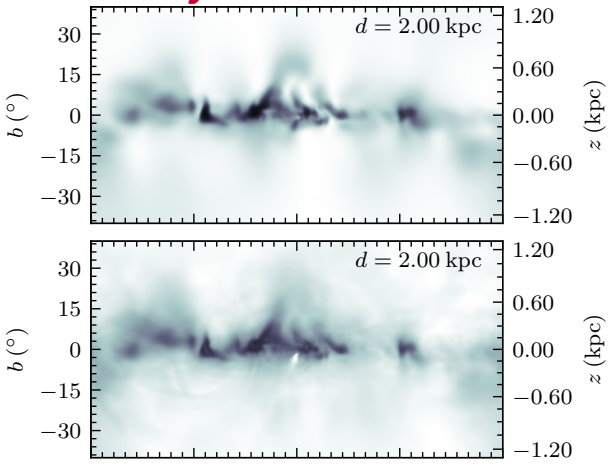
ϵ (top) vs. $\epsilon_{\rm L22}$ (bottom) at $d = 2$ kpc.

Putney et al. 2024

Validating the learned dust efficiency

Independent check: $\epsilon_{\rm L22}$ from the Lallement+22 3D dust map

- Red clump returns above the completeness cut
- Same clouds, same shells at 1, 2, 3 kpc



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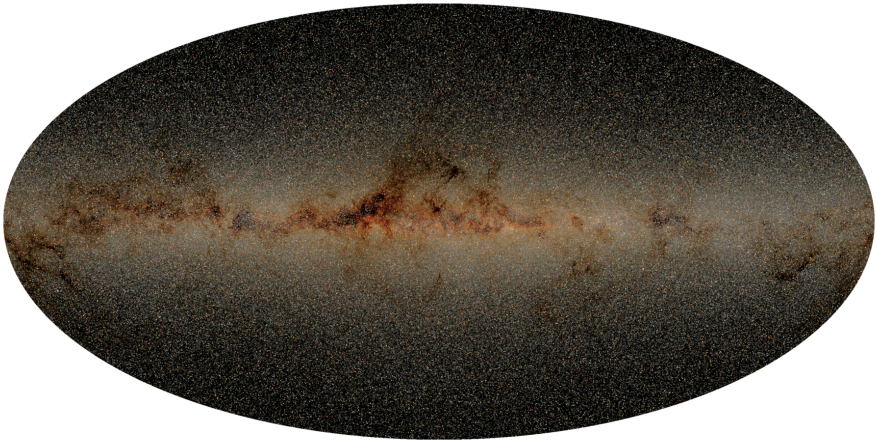
Putney et al. 2024

The dust-corrected Milky Way



Gaia DR3 tracers on the sky as observed: dust lanes carve the midplane. *Putney et al. 2024*

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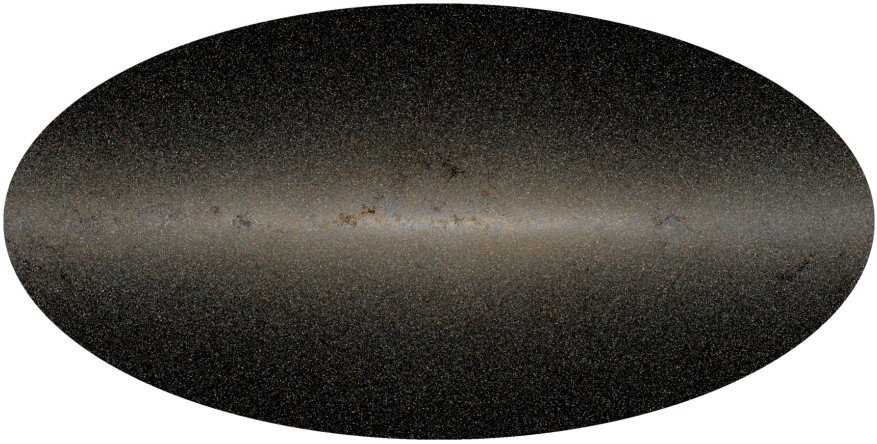
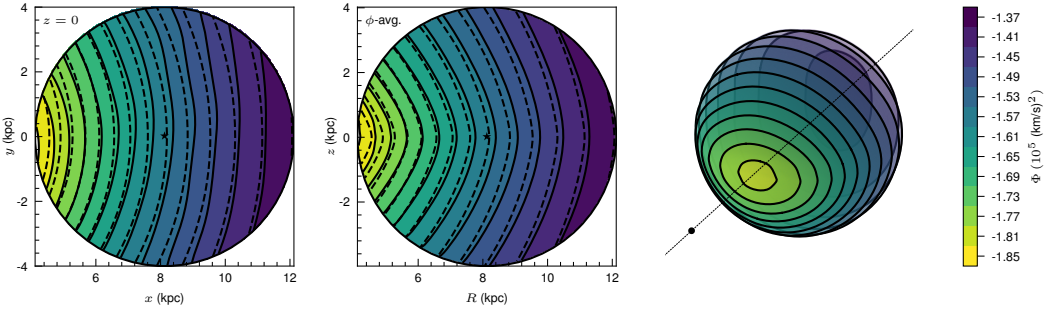


The same stars, reweighted by $1/\epsilon$ and de-reddened. *Putney et al. 2024*

The potential and the acceleration field

$\Phi(\vec{x})$ over the 4 kpc volume, smooth and dust-corrected.

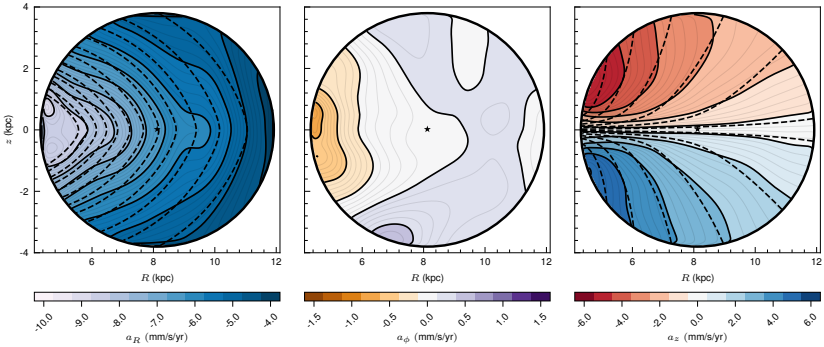


Φ: this work (solid) vs. MWPotential2014 (dashed): midplane, R–z, and isosurfaces around the Sun ($\star$).

Putney et al. 2025

The potential and the acceleration field

$\vec{a} = -\vec{\nabla}\Phi$: $a_\phi \neq 0$ at 3–5σ in the disk, so the field is not axisymmetric.



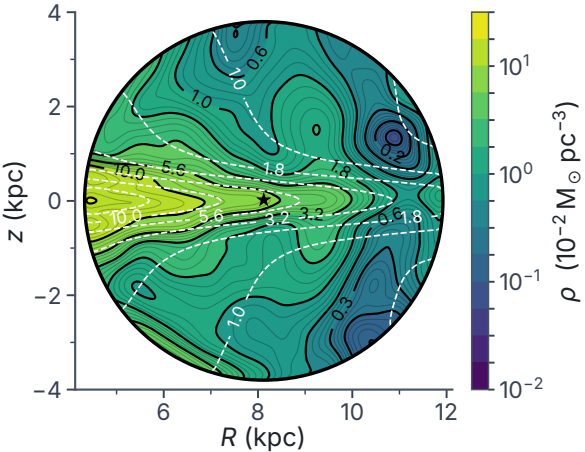
Left to right a_R, a_ϕ, a_z; dashed: MWPotential2014, which has $a_\phi \equiv 0$.

Putney et al. 2025

Mass density of the local Milky Way

$\rho = \nabla^2\Phi/4\pi G$ across the 4 kpc window:

- Stellar + gas disk at $|z| < 0.5$ kpc
- Non-zero density in the halo: DM
- Dip at $R \approx 11$ kpc, void above it



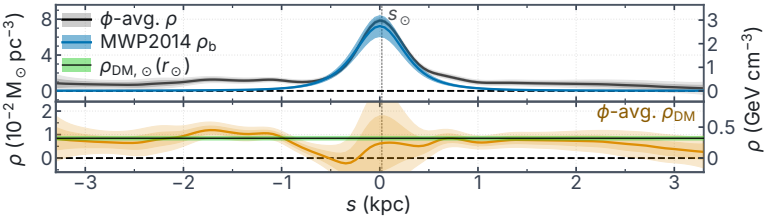
$\rho(R, z)$ (black) vs. MWPotential2014 (white dashed). *Putney et al. 2025*

The local dark matter density

- Subtract a baryonic model (stars + gas), average over the sphere $r = r_\odot$
- Errors: *Gaia*, statistical, training; systematics: the baryonic model

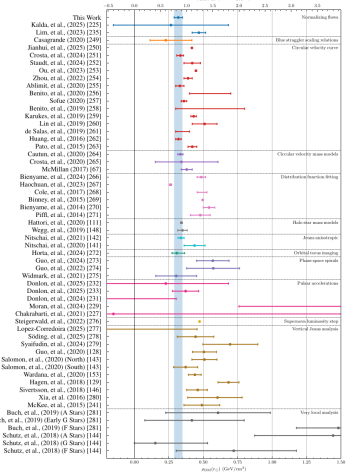
$$\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$$

spherical average at the Solar radius

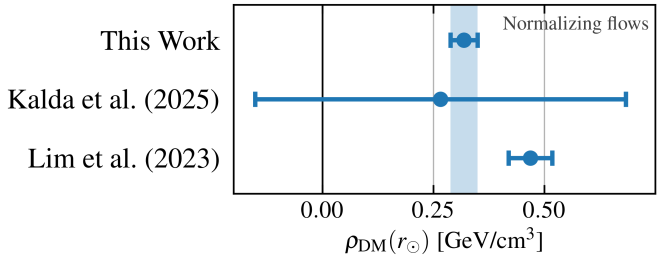


Total ρ and baryonic $\rho_{\rm b}$ (top); DM residual (bottom), along the arc through the Sun. *Putney et al. 2025*

The local dark matter density in context



One decade+ of $\rho_{\mathrm{DM},\odot}$.

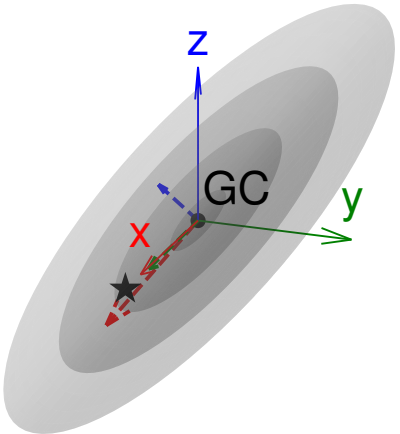


Putney et al. 2025

Halo shape

Tilted triaxial NFW fit:

- Axis ratios $b/a = 0.89 \pm 0.09$, $c/a = 0.33 \pm 0.06$: strongly oblate
- In-plane $\rho_{\mathrm{DM},\odot} = 0.43 \pm 0.03\,\mathrm{GeV/cm^3}$, 30% above the spherical average



Best-fit tilted oblate halo. *Putney et al. 2025*

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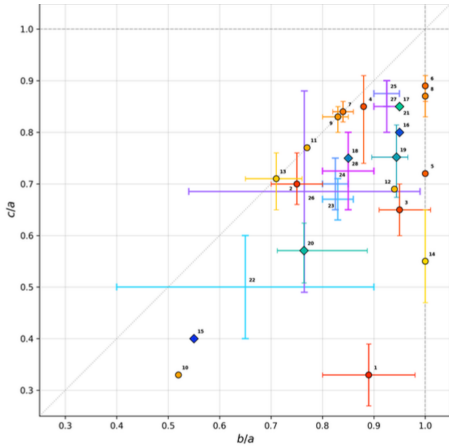

Preliminary. M. Scuderi, UNDARK 2 (Sept. 2026); Scuderi, Manconi & Cirelli, to appear.

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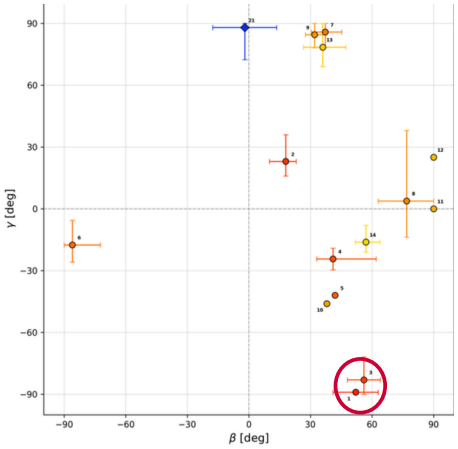
- Orientation agrees with a GD-1 stream fit [Nibauer+25] (points 1 and 3)



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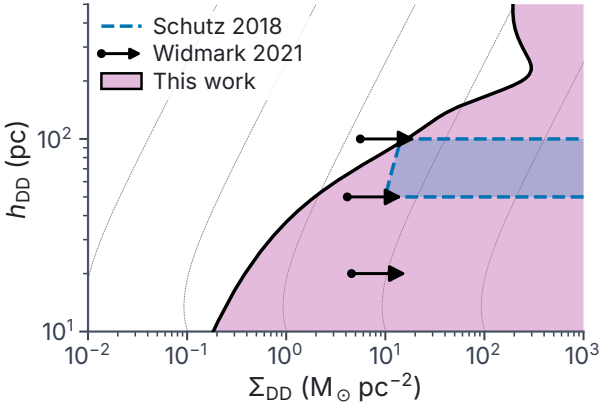
A dark disk?

Dissipative dark sector, a disk aligned with the baryons [Fan+13]:

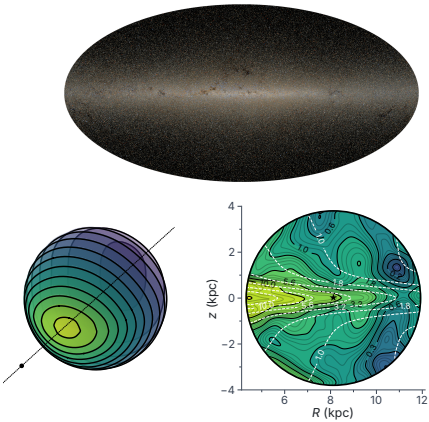
- Fit dark disk + NFW to the 3D $\rho_{\rm DM}$ field at $|z| < 500$ pc
- $10\times$ tighter limit on $\Sigma_{\rm DD}$ at $h_{\rm DD} = 20$ pc, $> 2\times$ at 50–100 pc



95% CL exclusion (pink) vs. Schutz+18, Widmark+21. *Putney et al. 2025*

To recap

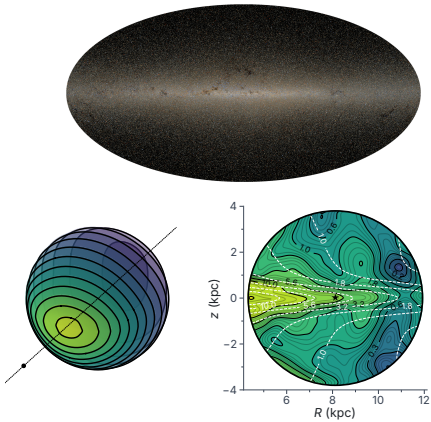
- Flows: the 6D density of 5.8 M *Gaia* stars
- PINN: Boltzmann equation $\rightarrow \Phi$ and dust ϵ, jointly
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$; oblate tilted halo; dark-disk limit



The dust-corrected sky, the potential, and the mass density, all from one fit.

To recap

- Flows: the 6D density of 5.8 M *Gaia* stars
- PINN: Boltzmann equation $\rightarrow \Phi$ and dust ϵ, jointly
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$; oblate tilted halo; dark-disk limit
- Next: *Gaia* DR4 (>100 M 6D stars)
- Approaching the Galactic Center!



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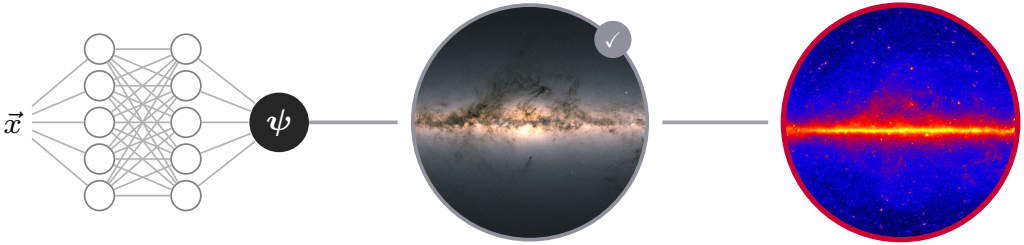
Roadmap



Physics-informed neural networks

The PDE residual as the loss function

Part I: ClearPotential

Dark matter from the kinematics of 5.8 M *Gaia* stars

with M. Buckley, S. H. Lim, D. Shih

Part II: PINNCR

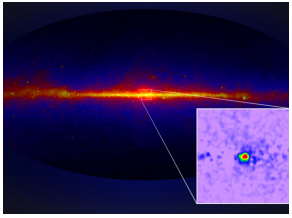
Diffuse γ-ray background to the GeV excess

with Y. Park, D. Shih, M. Buckley, T. Slatyer, C. Chen

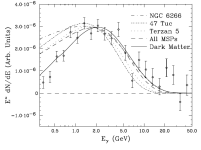
Images: ESA/Gaia/DPAC; NASA/DOE/Fermi-LAT Collaboration

The Galactic Center GeV excess

- GeV γ-ray excess from the inner Milky Way [Goodenough & Hooper 09]



Residual after the diffuse model. *Goodenough & Hooper (2009)*



The excess spectrum peaks near 2 GeV. *Daylan et al. (2014)*

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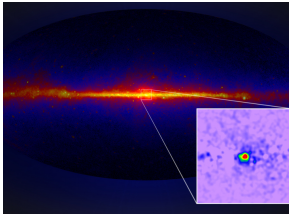
- Or $\gtrsim 10^4$ unresolved millisecond pulsars [Hooper & Mohlabeng 16]



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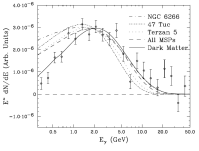

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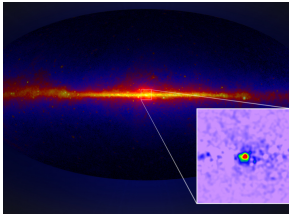
- Or $\gtrsim 10^4$ unresolved millisecond pulsars [Hooper & Mohlabeng 16]
- Point sources or smooth? Tests depend on the background model



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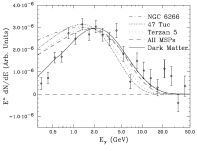

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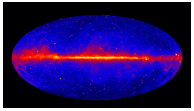
Where the background comes from

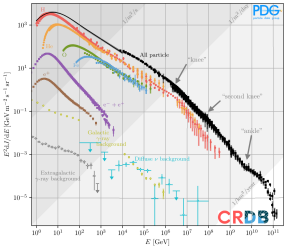
Diffuse γ-rays = cosmic rays (CRs) hitting interstellar gas and light:

- π^0 decay ($p \times$ gas)
- Inverse Compton ($e^- \times$ starlight, CMB)
- Bremsstrahlung ($e^- \times$ gas)

So we need CRs across the *whole* Galaxy, not just what we measure locally.



Fermi-LAT sky, > 1 GeV: mostly diffuse emission.
NASA / DOE / Fermi-LAT Collaboration



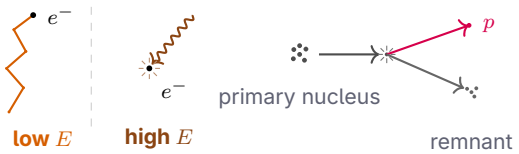
CR spectrum at Earth: one point in the Galaxy. *PDG, Review of Particle Physics 2025*

The cosmic-ray transport equation

The CR density $\psi_i(\vec{r}, p)$ of each species obeys the transport equation:

$$0 = q(\vec{r}, p) + \vec{\nabla} \cdot \left(D_{xx} \vec{\nabla} \psi - \vec{V} \psi \right) + \frac{\partial}{\partial p} p^2 D_{pp} \frac{\partial}{\partial p} \frac{1}{p^2} \psi - \frac{\partial}{\partial p} \left[\dot{p} \psi - \frac{p}{3} (\vec{\nabla} \cdot \vec{V}) \psi \right] - \frac{\psi}{\tau_f} - \frac{\psi}{\tau_r}$$

- Source, diffusion, convection, reacceleration, energy losses, fragmentation, decay.
- Species are coupled.
- Dozens of transport parameters $\vec{\eta}$ (D_0, δ, z_L, injection, ...).

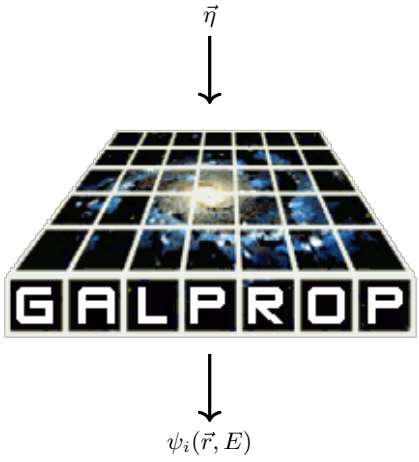


Diffusion, radiative loss, fragmentation.

GALPROP

Standard numerical solver code:

- Pick $\vec{\eta}$, I.C., evolve ψ_i to steady state on a grid.
- CPU cost $\propto N_x^3\, N_E\, N_{\rm species}\, N_{\rm steps}$.
- High res. ≈ 100 CPU-h, memory intensive – *per* $\vec{\eta}$.
- A 10^4-run scan: $\gtrsim$15 CPU-yr, even in 2D.



[Strong & Moskalenko 98; Porter+22]

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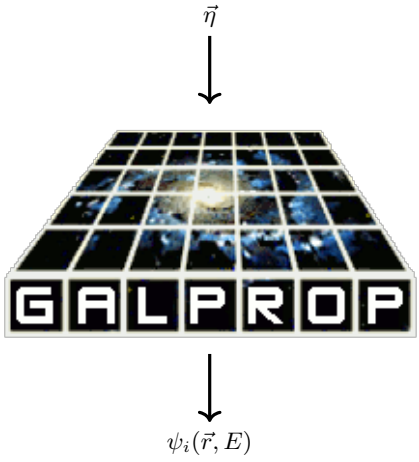
Need a surrogate that returns ψ_i in seconds.

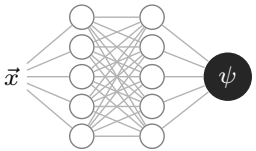


[Strong & Moskalenko 98; Porter+22]

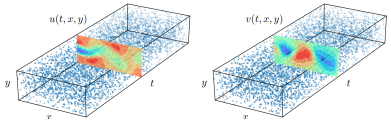
Physics-informed neural networks

Solve a PDE with a neural network instead of a grid:

- The solution is a network, $\psi_\theta(\vec{x})$: smooth and differentiable by construction



$\psi_\theta(\vec{x})$ as a network; the PDE residual is the loss.



Navier–Stokes. *Raissi et al., arXiv:1711.10566*

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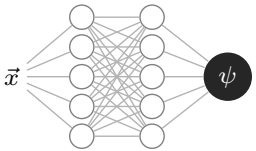

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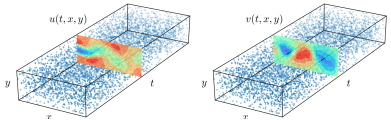

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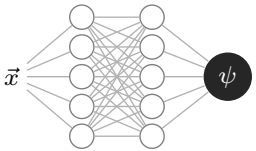
- The loss is the PDE residual, $\mathcal{L}_\theta = \left\langle |\mathcal{D}[\psi_\theta]|^2 \right\rangle_{\vec{x}}$, with derivatives by autograd
- No grid: random points in the domain, boundary conditions built in or added to the loss



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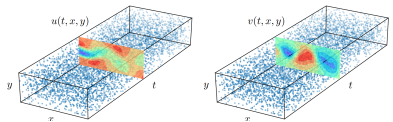

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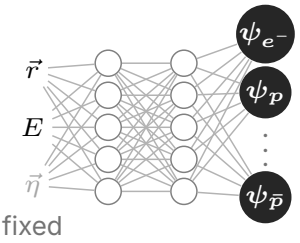
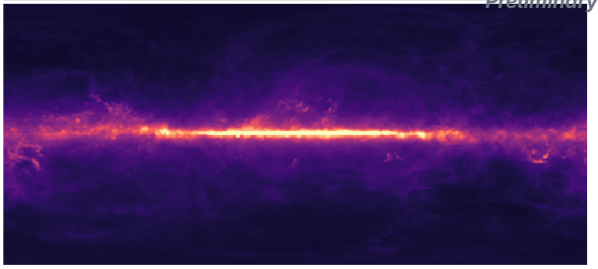
PINNs for cosmic rays and diffuse emission

1. CR flux as a PINN: $\vec{\psi}_\theta(\vec{x}, E)$. Same recipe, now for cosmic rays: the transport-equation residual is the loss.





PINN γ-ray sky at 5 GeV: one model (C), then six (A–F).

PINNs for cosmic rays and diffuse emission

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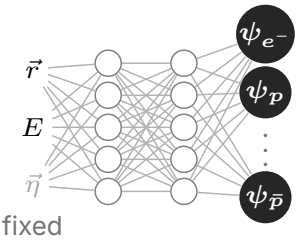
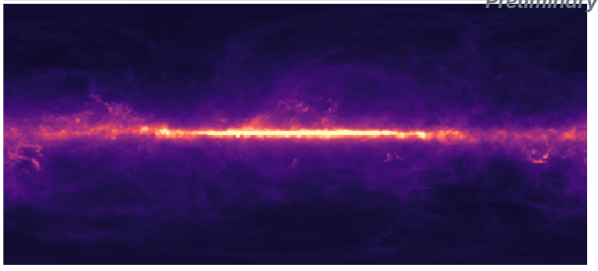
2. Add $\vec{\eta}$ as an input: $\vec{\psi}_\theta(\vec{x}, E, \vec{\eta})$. Fixed $\vec{\eta}$: one GALPROP run.





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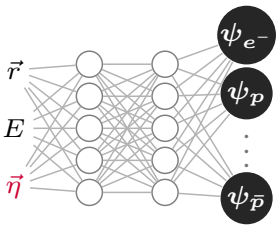
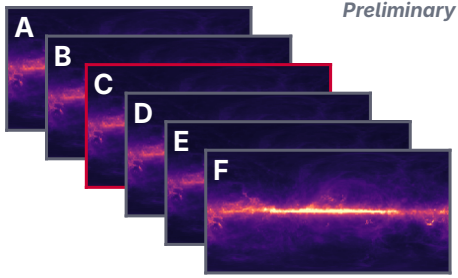


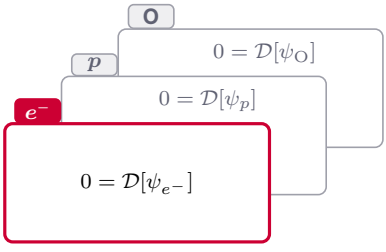


PINN γ-ray sky at 5 GeV: one model (C), then six (A–F).

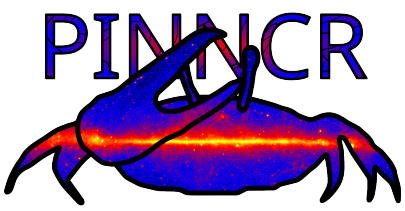
PINNCR

Differentiable Python code that assembles the transport equation and trains a PINN to solve it:

- **Input:** CR species $\{\psi_i\}$ with their processes and couplings
- **Output:** one network trained on every species' transport residual



$$\mathcal{L} = \frac{1}{N} \sum_i \sum_s w_s \left| \mathcal{D}_s[\psi](\vec{r}_i, E_i, \vec{\eta}) \right|^2$$



Training PINNCR

Standard and non-standard tricks to get PINNCR to converge:

PDE non-dimensionalization

- fields and coefficients scaled to $\mathcal{O}(1)$
- flux in $\log\psi$: 8 decades become one
- log-ratio residual

Network architecture

- boundary built in: $\psi_\theta = B(\vec{r})\,\mathcal{N}_\theta$, $B = 0$ at the halo edge
- factorized: $\log\psi = \log\phi(\vec{r}, E) + \log S(E)$

Training algorithm

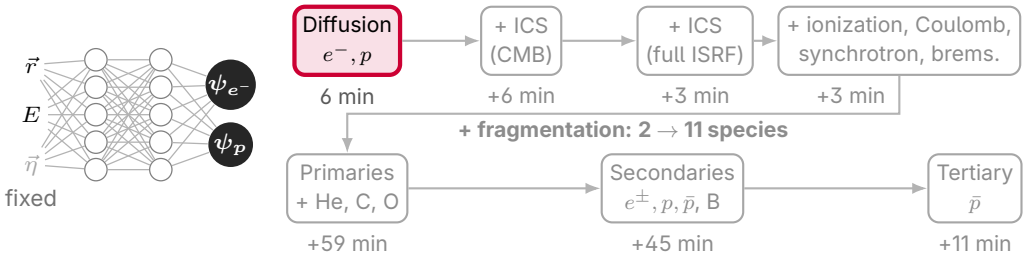
- curriculum: add one process at a time

Structured after Wang, Sankaran & Perdikaris, arXiv:2308.08468.

Curriculum learning

Start with simple physics, add processes one at a time.

$$0 = q(\vec{r},p) + \vec{\nabla}\cdot\left(D_{xx}\vec{\nabla}\psi\right) - \vec{\nabla}\cdot(\vec{V}\psi) + \frac{\partial}{\partial p}p^2 D_{pp}\frac{\partial}{\partial p}\frac{1}{p^2}\psi - \frac{\partial}{\partial p}\left[\dot{p}\,\psi - \tfrac{p}{3}(\vec{\nabla}\cdot\vec{V})\psi\right] - \frac{\psi}{\tau_f} - \frac{\psi}{\tau_r}$$

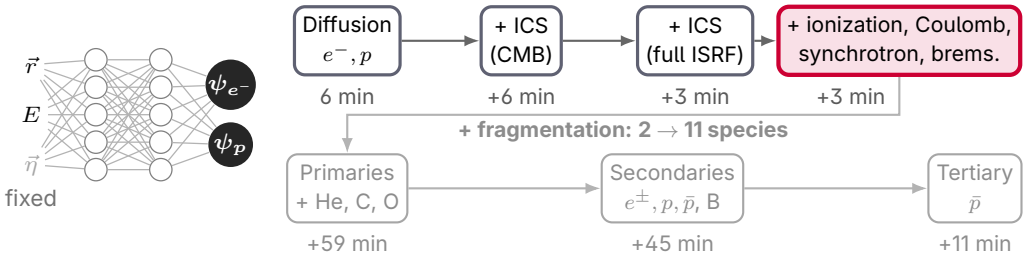


Training time per stage: median over seeds, one GPU.

Curriculum learning

Start with simple physics, add processes one at a time.

$$0 = q(\vec{r}, p) + \vec{\nabla}\cdot\left(D_{xx}\vec{\nabla}\psi\right) - \vec{\nabla}\cdot(\vec{V}\psi) + \frac{\partial}{\partial p}p^2 D_{pp}\frac{\partial}{\partial p}\frac{1}{p^2}\psi - \frac{\partial}{\partial p}\left[\dot{p}\,\psi - \tfrac{p}{3}(\vec{\nabla}\cdot\vec{V})\psi\right] - \frac{\psi}{\tau_f} - \frac{\psi}{\tau_r}$$



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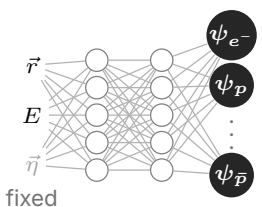
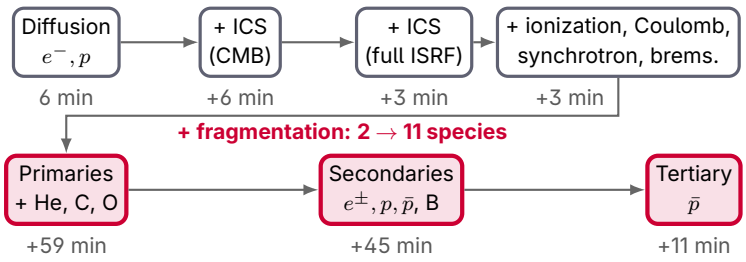
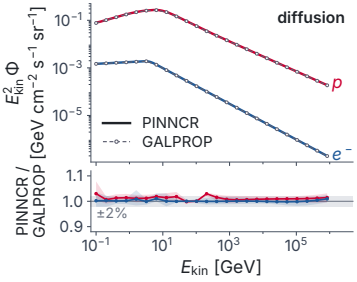




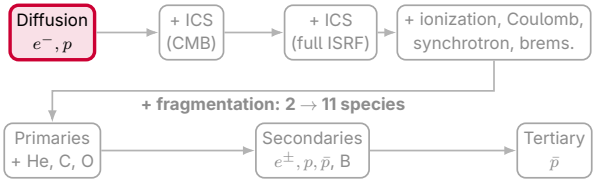
Training time per stage: median over seeds, one GPU.

Diffusion, then energy losses

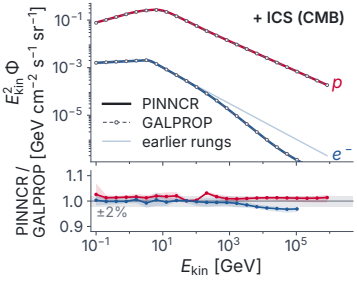


Diffusion + source: e^- and p at the Sun.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.

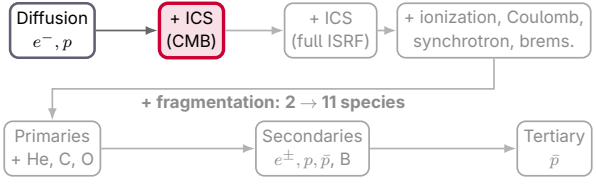


Diffusion, then energy losses

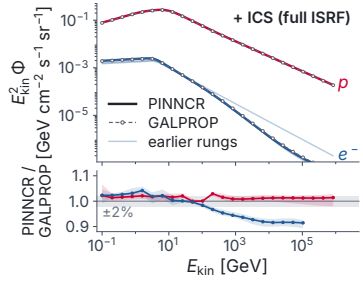


+ inverse Compton on the CMB. Faded: earlier rung.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.

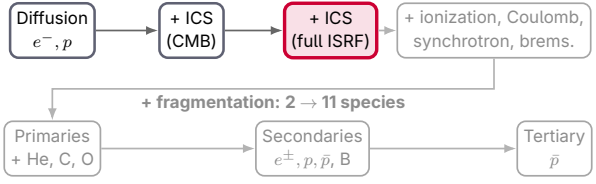


Diffusion, then energy losses

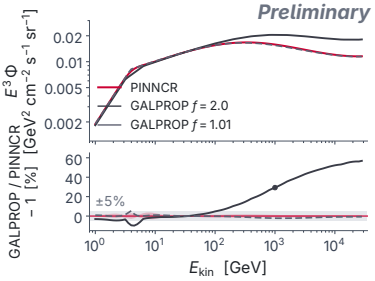


- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.
- + full ISRF: e^- ratio 0.92–0.95 at 1–100 TeV, under study.

+ inverse Compton on the full ISRF (optical + IR).

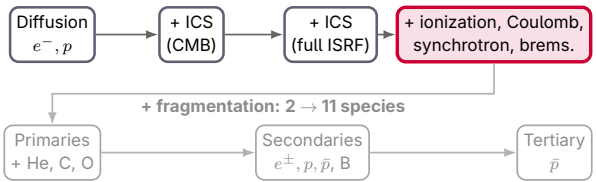


Diffusion, then energy losses

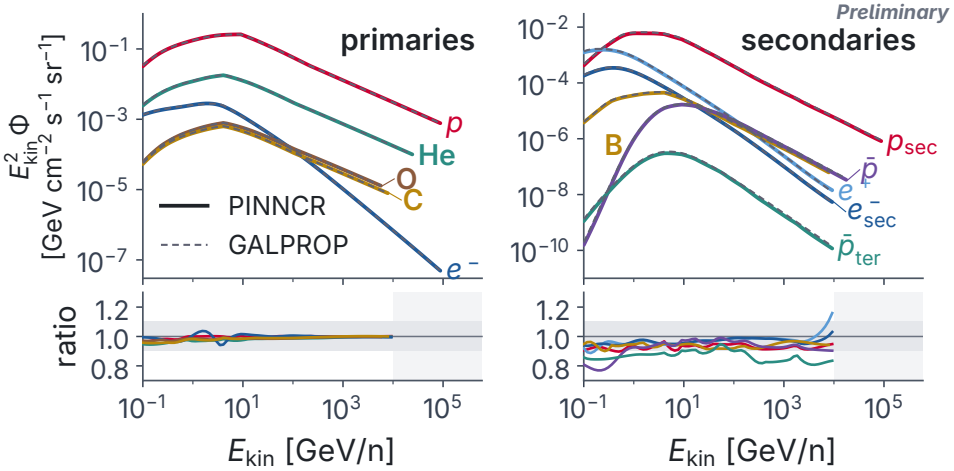


+ ICS and losses: TeV e^- vs. GALPROP's energy grid, $f = E_{i+1}/E_i$.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.
- + full ISRF: e^- ratio 0.92–0.95 at 1–100 TeV, under study.
- + ICS: GALPROP's default grid is 29% off at 1 TeV; matching PINNCR costs ∼70× the E-bins.

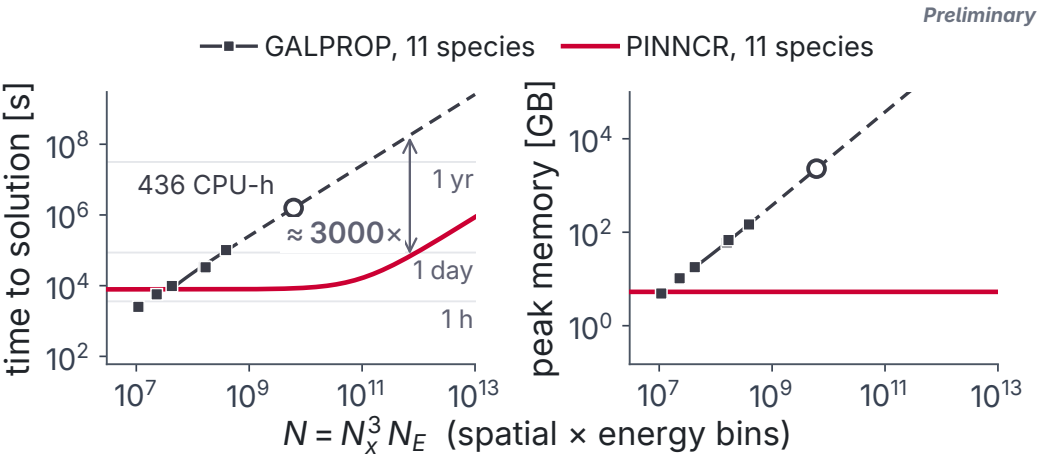


Full-physics GALPROP surrogate model



Same $Z \leq 8$ nuclear network in both codes. Ratio panels: grey band $\pm 10\%$, shaded energies beyond the training range.

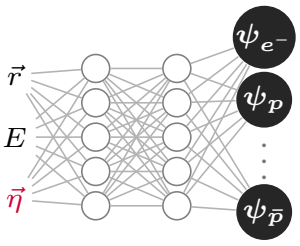
Cost comparison

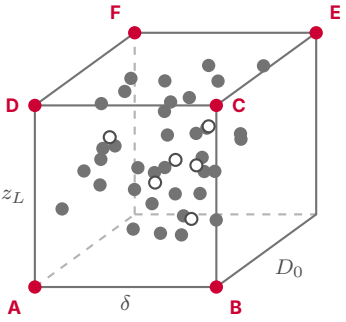


Time to solution and peak memory vs. grid size, 11 species: GALPROP on CPU (squares measured, dashed extrapolated, circle the production run) vs. PINNCR on one GPU.

Varying the transport parameters $\vec{\eta}$

Let $\vec{\eta}$ vary – one network covers a hypercube of diffuse models.



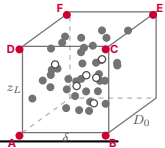


$\vec{\eta}$: 15 parameters (3 axes shown)

- training points (uniform)
- Cholis+21 models A–F
- held-out tests

Fiducial diffuse models

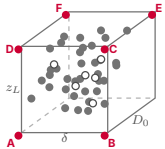
Six published GALPROP models set the training box, each axis padded by 8%.



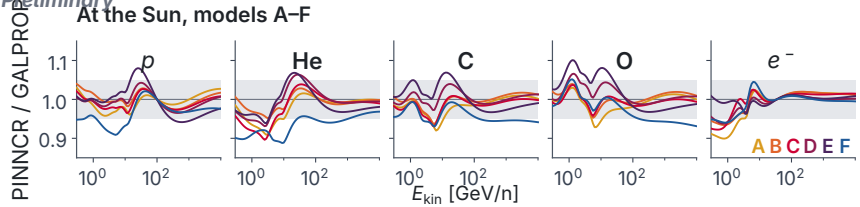
Model	δ	z_L [kpc]	D_0 [10^{28} cm^2/s]	v_A [km/s]	dv_c/dz [km/s/kpc]	injection, H / He γ_1	$R_{\mathrm{br},1}$ [GV]	γ_2	$R_{\mathrm{br},2}$ [GV]	γ_3
A	0.33	5.7	6.70	30.0	0	1.74/1.70	6.0/7.4	2.04/2.16	14.0/21.5	2.41/2.39
B	0.37	5.5	5.50	30.0	2	1.72/1.74	6.0/8.0	2.00/2.14	12.4/21.0	2.38/2.375
C	0.40	5.6	4.85	24.0	1	1.69/1.65	6.0/6.7	2.00/2.13	12.4/20.0	2.38/2.355
D	0.45	5.7	3.90	24.0	5.5	1.69/1.68	6.0/7.0	1.99/2.12	12.4/18.7	2.355/2.34
E	0.50	6.0	3.10	23.0	9	1.71/1.68	6.0/7.2	2.02/2.14	11.2/17.5	2.38/2.33
F	0.43	3.0	1.85	20.0	2	1.68/1.74	6.0/10.5	2.08/2.09	13.0/21.0	2.41/2.33

Cholis et al. (2021), Table I

Accuracy across models A–F



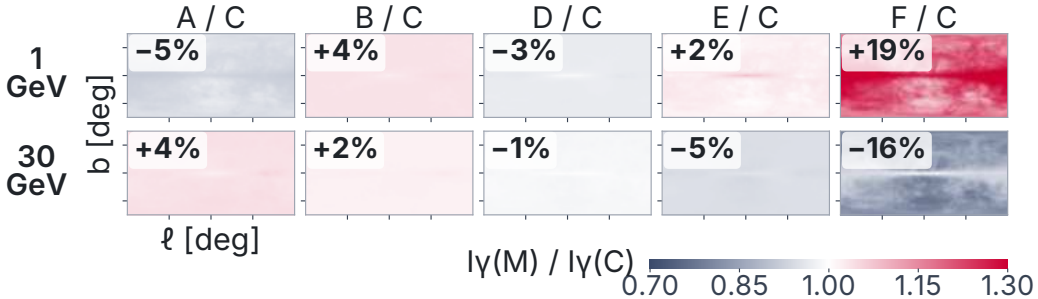
Preliminary



Grey band: ±5%. Medians 1–5% at the Sun (A–F); every disk pixel within 5% at 10 GeV/n (C).

The sky across the fiducial models

Preliminary

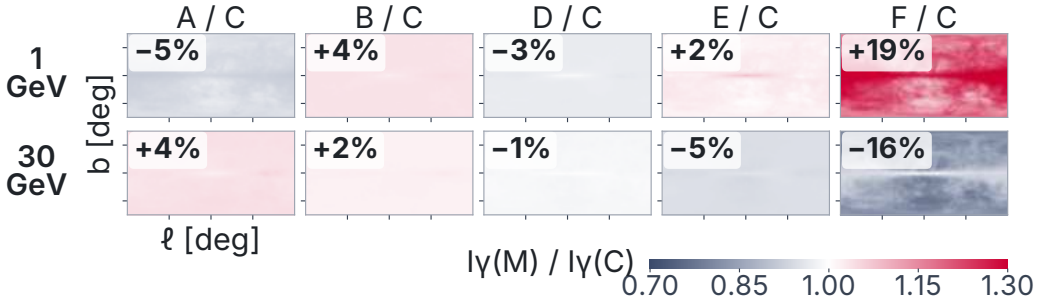


Sky ratio to model C for Cholis+21 models A–F (top 1 GeV, bottom 30 GeV), one PINNCR network.

- Models A–F change the sky by up to 19%.

The sky across the fiducial models

Preliminary

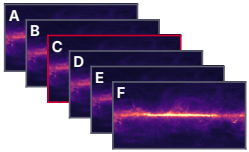


Sky ratio to model C for Cholis+21 models A–F (top 1 GeV, bottom 30 GeV), one PINNCR network.

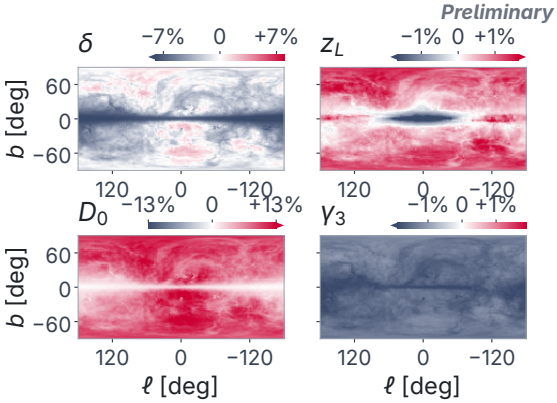
- Models A–F change the sky by up to 19%.
- Now: bracket the excess with six models. Next: a *continuous* posterior over transport.

Differentiable γ-ray maps

- Integrate ψ against gas + ISRF: a γ-ray sky in minutes per $\vec{\eta}$.
- Derivatives $\partial(\text{sky})/\partial\vec{\eta}$: one backward pass.
- Likelihood gradients in $\vec{\eta}$ cost one backward pass: which transport directions mimic the excess?



PINN γ sky, 5 GeV, models A–F.

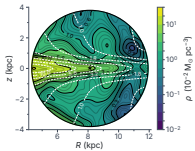


% change of I_γ per transport parameter, model C, 30 GeV.

Summary

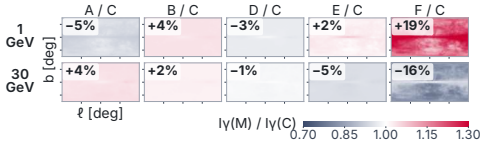
✓ Part I: ClearPotential

- Flows + a PINN: f, Φ and ϵ from 5.8 M *Gaia* stars
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$
- Oblate tilted halo, dark-disk limit



✓ Part II: PINNCR

- CR transport, 11 species, GALPROP accuracy, no grid
- One network for different η; γ sky + gradients in minutes
- Next: the transport-marginalized GeV excess fit

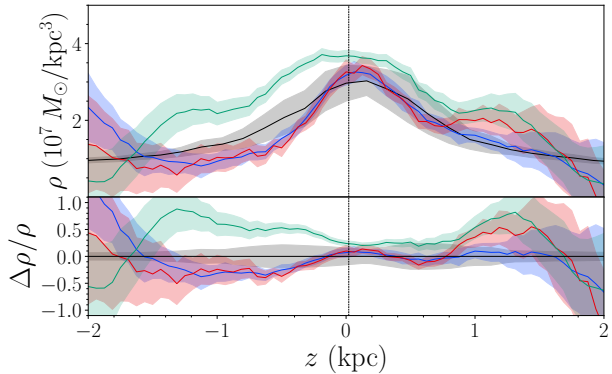


Thanks for listening!

`arXiv:2412.14236`, `2512.09989` (ClearPotential) · `arXiv:2610.XXXXX` + GitHub (PINNCR)

Backup: Validation on the h277 simulation

- N-body + SPH zoom simulation of a MW analog (GASOLINE); truth-level Φ, $\vec{a}$, ρ [Zolotov+12]
- Mock *Gaia* observation: $153{,}714$ star particles within 3.5 kpc, *Gaia*-like errors
- Flows recover the bulk PSD to $\lesssim$1%; pointwise CBE gives $\vec{a}$ to 1–5%
- ρ within $\mathcal{O}(20\%)$; a Jeans analysis is biased high by $\mathcal{O}(50\%)$



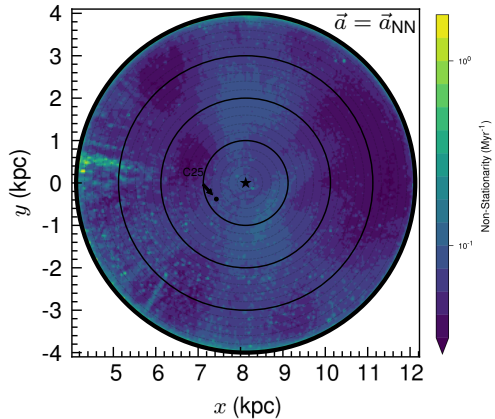
ρ along the z-axis at $x_\odot$: flow vs. Jeans vs. truth. *Buckley et al. 2022*

Backup: A first map of non-stationarity in the Milky Way

Average the CBE residual over velocity:

$$\mathrm{NS}^2(\vec{x}) \equiv \mathbb{E}_{\vec{v}}\left[\mathrm{CBE}^2(\vec{x}, \vec{v} \,|\, \vec{a})\right] \quad [\mathrm{Myr}^{-2}]$$

- $\mathrm{NS} = 0$ in equilibrium; grows with disequilibrium or mismodelling
- Traces the spiral arms in the h277 simulation



NS in the x–y plane with the PINN acceleration $\vec{a}_{\mathrm{NN}}$; dot: Chandra+25 subhalo.

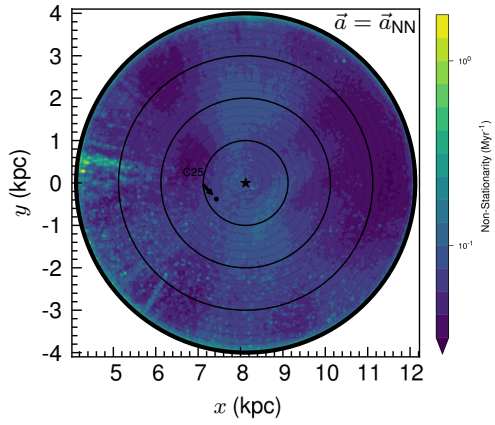
Putney et al. 2025

Backup: A first map of non-stationarity in the Milky Way

Average the CBE residual over velocity:

$$\mathrm{NS}^2(\vec{x}) \equiv \mathbb{E}_{\vec{v}}\left[\mathrm{CBE}^2(\vec{x}, \vec{v} \,|\, \vec{a})\right] \quad [\mathrm{Myr}^{-2}]$$

- $\mathrm{NS} = 0$ in equilibrium; grows with disequilibrium or mismodelling
- Traces the spiral arms in the h277 simulation
- First map across 4 kpc: a possible arm at $R_\odot$, nothing at the Chandra+25 subhalo



NS in the x–y plane with the PINN acceleration $\vec{a}_{\mathrm{NN}}$; dot: Chandra+25 subhalo.

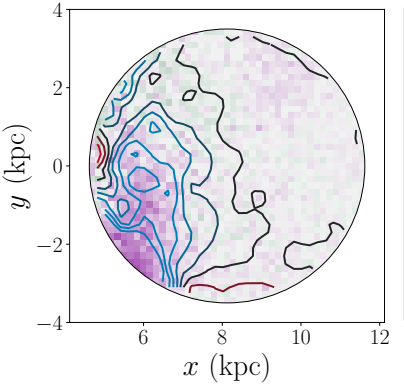
Putney et al. 2025

Backup: Non-stationarity in the simulation

Use the truth-level $\vec{a}$, integrate over v:

$$\mathrm{NS}^2(\vec{x}|\vec{a}_{\mathrm{sim}}) \equiv \mathbb{E}_{\vec{v}}[\mathrm{CBE}^2(\vec{x},\vec{v}|\vec{a}_{\mathrm{sim}})]$$

- $\mathrm{NS} = 0$ if equilibrium holds; measures the local disequilibrium timescale
- NS contours trace the spiral arms (overdensities $\Delta\rho_\star$)



Baryonic over/underdensities (purple/green) and the NS proxy (red/blue).

Buckley et al. 2022

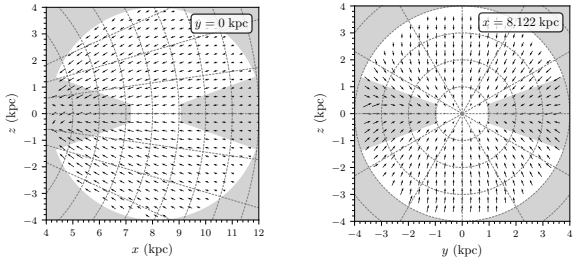
Backup: 2023 pointwise analysis, and why $\rho_{\rm DM}$ moved

Pointwise CBE at every $\vec{x}$ across the 4 kpc volume [Lim+23]:

- $|\vec{a}_\odot| = 1.93 \pm 0.08 \times 10^{-10}$ m/s^2, $< 2\sigma$ from the EDR3 quasar measurement
- $\rho_{\rm DM}(r_\odot) = 0.47 \pm 0.05\,{\rm GeV/cm^3}$ from 15 points at $r_\odot$

ClearPotential (0.32 ± 0.03) is $\sim 3\sigma$ lower. Why:

- **Dust**: ϵ corrects 14 of the 15 "dust-free" points by $\sim$10%
- **Curl-free** $\vec{a} = -\vec{\nabla}\Phi$ instead of independent pointwise $\vec{a}$
- 215 points over the sphere, dozens inside the disk, vs. 15 outside it

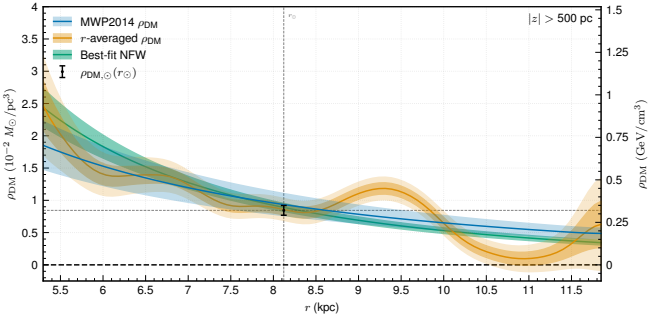


$\vec{a}$ in the xz plane ($y = 0$, side view) and the yz plane ($x = x_\odot$, facing the GC). *Lim et al. 2023*

Backup: Halo profile

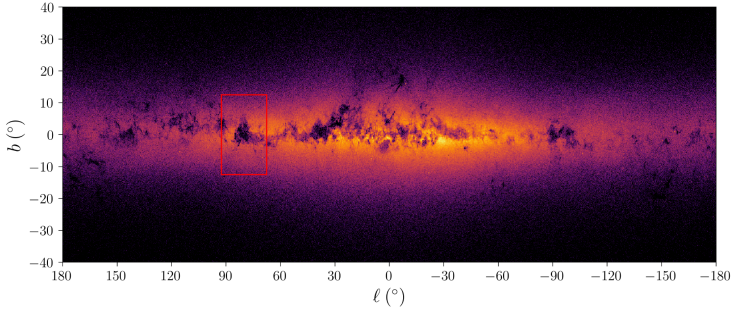
Profile fits using $\rho_{\rm DM}$ measured across the entire volume:

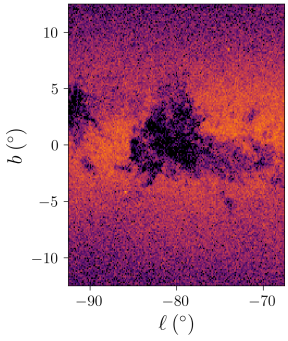
- Highly degenerate over $4 < r < 12$ kpc; systematics dominated by the **baryonic model**
- Weak inner-profile sensitivity
- NFW ($\beta = 1$): $\rho_{\rm DM,\odot} = 0.34^{+0.02}_{-0.02}$ GeV/cm^3, $r_s = 3.1^{+2.2}_{-1.5}$ kpc
- gNFW: short $r_s = 1.9^{+1.3}_{-1.0}$ kpc, flat $\beta = 0.0^{+1.4}_{-0.0}$



Spherically averaged $\rho_{\rm DM}(r)$ (orange) vs. MWPotential2014 (blue) and best-fit NFW (green); crosshair: $r_\odot$, $\rho_{\rm DM,\odot}$. *Putney et al. 2025*

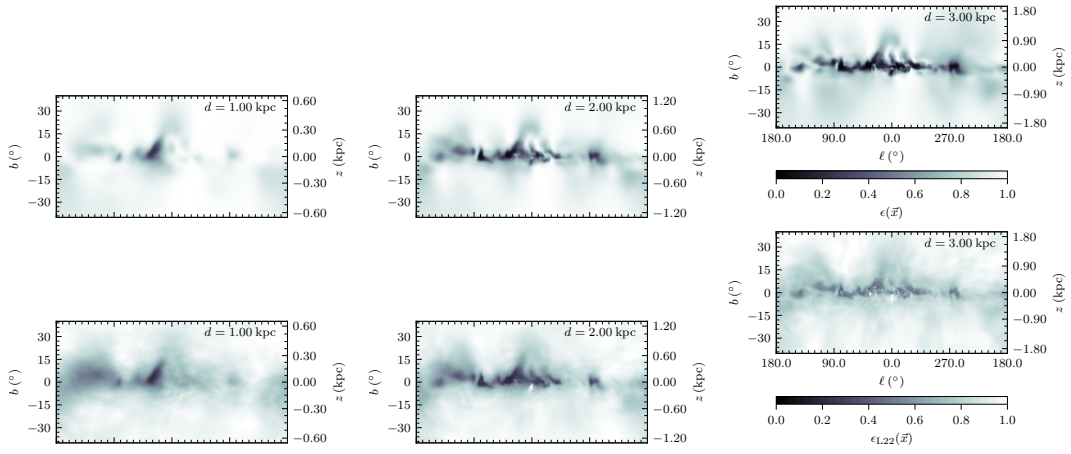
Backup: Dust lanes in the *Gaia* sample





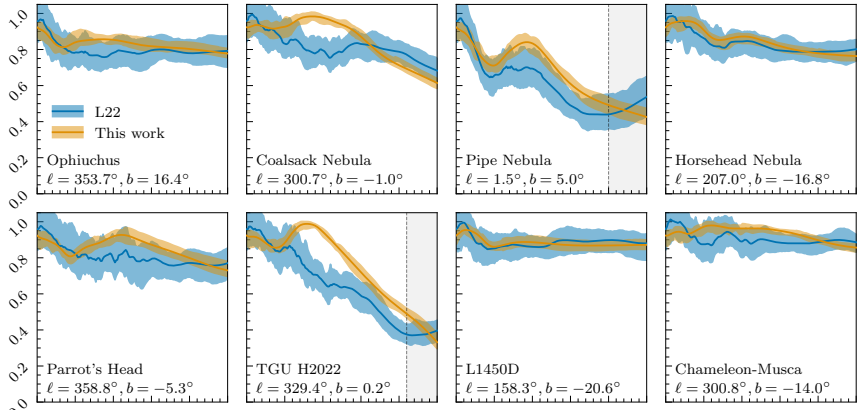
Gaia RC + RGB stars on the sky; the Vulpecula rift (box, zoom at right) is a hole in the sample. *Putney et al. 2024*

Backup: ϵ vs. $\epsilon_{\rm L22}$ at 1, 2, 3 kpc



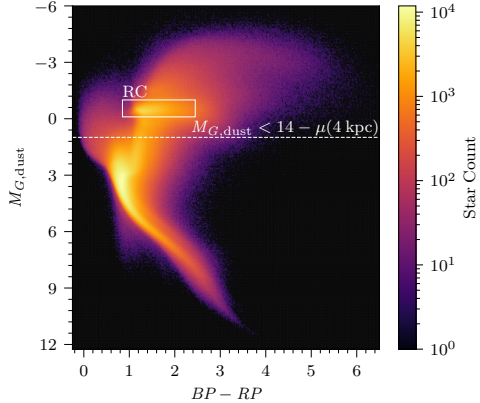
ϵ (top) vs. $\epsilon_{\rm L22}$ (bottom) at $d = 1, 2, 3$ kpc. *Putney et al. 2024*

Backup: ϵ toward known dust clouds



ϵ (gold) vs. ϵ_{L22} (blue) along $10'$ cones toward named clouds; bands: meas. + stat. errors. *Putney et al. 2024*

Backup: Dust correction restores the red clump

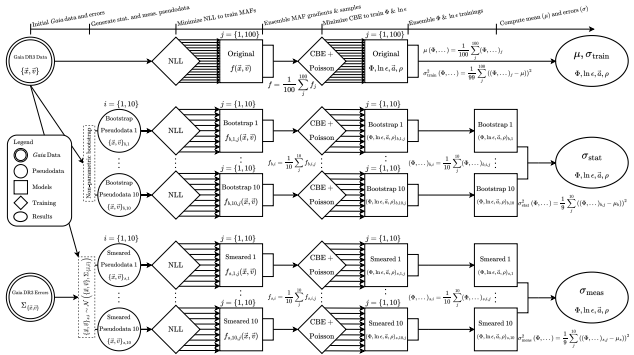


CMD, $d < 4$ kpc, L22-corrected: the red clump lifts above the completeness cut. *Putney et al. 2024*

Backup: Pipeline and error budget

Three error sources, propagated by re-training:

- **Measurement**: 10 catalogs re-drawn from DR3 covariances; 10 flows + 10 nets each
- **Statistical**: 10 bootstrap resamples, same treatment
- **Training**: 100 seeds; spread/$\sqrt{N}$, subdominant
- **Systematic** ($\rho_{\rm DM}$ only): baryonic model, McKee+17 vs. MWPotential2014



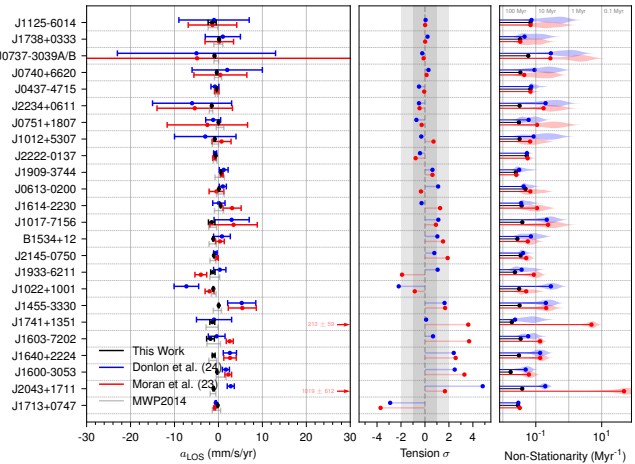
From *Gaia* data to ϵ and Φ: 300 (Φ, ϵ) networks in all; 2-pt correlations of ρ kept in every average and fit.

Putney et al. 2024

Backup: Non-stationarity from binary pulsar accelerations

Compare $\vec{a}_{\rm LOS}$ to 24 binary pulsars [Donlon+24, Moran+24], independent of equilibrium:

- $17/24$ within 2σ; three in slight tension
- Avg $t_{\rm dyn} = 28.9$ Myr (us) vs. $9.9/6.1$ Myr (Donlon/Moran)
- Independent accelerations can map real non-stationarity



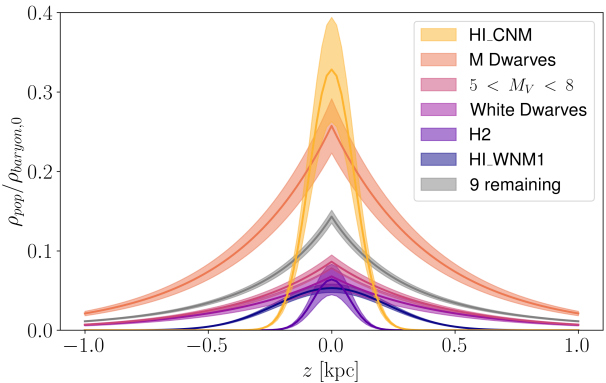
LOS $\vec{a}$, tension vs. this work, and NS per pulsar; Donlon blue, Moran red. *Putney et al. 2025*

Backup: Baryonic mass models

DM density = total ρ − baryonic model.
Two adopted models:

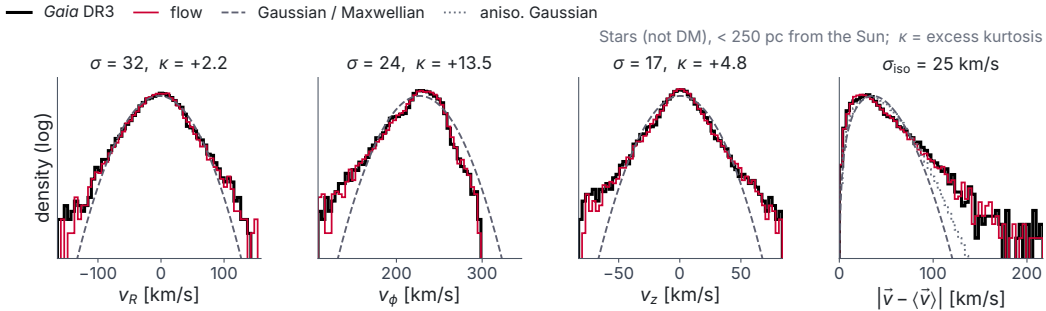
- **McKee 2017**: 15 components, 10 stellar (thin/thick disk, bulge, halo) + 5 gas (H_2, H I, ionized)
- **MWPotential2014** (Bovy 2015, galpy): compact 5-component analytic

Each component: exponential in R, exponential or Gaussian in $|z|$. **All assume axisymmetry.**



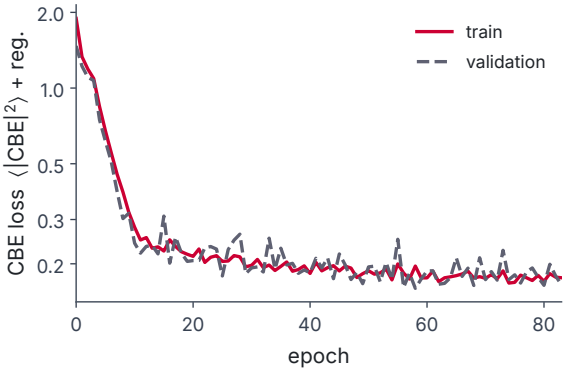
Lim et al. 2023

Backup: Local stellar velocities



Backup: Training the Φ / $\ln \epsilon$ PINN

- Two MLPs, $3 \to 100 \times 5 \to 1$, smooth (GINT) activations, one each for Φ and $\ln \epsilon$
- Inputs: flow gradients $\vec{\nabla}_x \ln f$, $\vec{\nabla}_v \ln f$ pre-sampled at $(\vec{x}, \vec{v})$
- Loss: MSE of the CBE residual, plus $\langle \ln \epsilon \rangle = 0$ and a positive-density penalty
- Ensembles: 100 seeds, plus bootstrap and re-perturbed catalogs for the error model



One seed, original catalog: train and validation loss per epoch.

Backup: CBE loss in detail

Pointwise [Buckley+22, Lim+23]:

$$\mathcal{L}_{\rm MSE}(\vec{x}) = \tfrac{1}{N}\sum_{\beta=1}^{N} \left| v_i \partial_{x_i} f + a_i \partial_{v_i} f \right|^2_{\vec{v}_\beta}$$

Set $\partial\mathcal{L}/\partial a_i = 0$: a 3×3 linear system at each $\vec{x}$,

$$V_i + M_{ij}\, a_j = 0,$$

$$M_{ij} = \sum_\beta \partial_{v_i} f\, \partial_{v_j} f, \quad V_i = \sum_\beta \left(v_j \partial_{x_j} f\right) \partial_{v_i} f$$

Functional [Putney+25]:

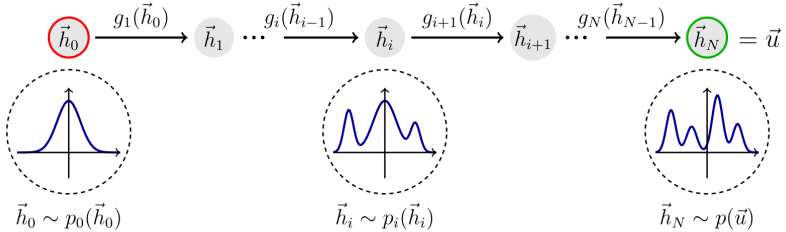
$$\mathcal{L}_\theta = \sum_{\vec{x}_i, \vec{v}_j} \left| \vec{v} \cdot \vec{\nabla}_x \ln f - \vec{\nabla}\Phi_\theta \cdot \vec{\nabla}_v \ln f \right|^2$$

- $\Phi_\theta(\vec{x})$ = NN; minimize over θ via SGD
- Regularize: smoothness, physical priors
- Reject $|\vec{v}| > 0.8\, v_{\rm max}$: no flow-tail blow-ups

Backup: Masked autoregressive flows

Factorize the joint by the chain rule, $p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_{<i})$; each conditional is a Gaussian with NN-learned (μ_i, σ_i).

- **Masking** (MADE): one MLP outputs all $\{(\mu_i, \sigma_i)\}$ at once; weights masked so $x_i \not\to (\mu_i, \sigma_i)$
- Stack several MADE blocks with permutations between them
- Exact log-likelihood, exact Jacobian, cheap gradients in $\vec{x}$ and $\vec{v}$

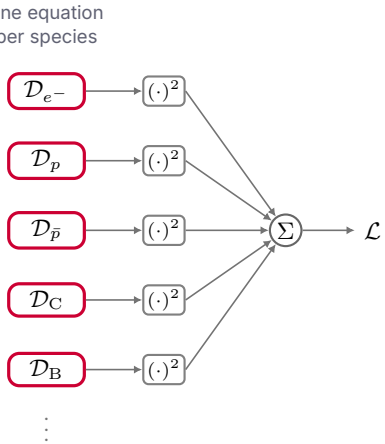


Flow diagram: thesis ch. 3

Backup: Why not an emulator trained on GALPROP runs?

- An emulator needs the training set first: 10^3–10^4 GALPROP runs is the cost we are trying to avoid, and it grows with the number of $\vec{\eta}$ axes.
- It inherits the solver's grid and its energy-grid bias; the PINN has no grid and matched GALPROP's fine-grid limit where the default grid was $\sim$9% high at 1 TeV.
- Between training points an emulator interpolates; the PINN residual $\mathcal{D}[\psi_\theta]$ can be evaluated at any $(\vec{r}, E, \vec{\eta})$ as a check.
- Cost side: the PINN's up-front training is $\sim$90 GPU-h for the 15-axis cube (10 seeds), about 5–6 GALPROP runs at the 3 MeV reference card (16 CPU-h each).
- Where an emulator wins: a smooth, low-dimensional map with a solver you trust and already ran.

Backup: PINNCR: one loss for every species



Each species' transport equation, written as $\mathcal{D}_s[\psi] = 0$:

$$\mathcal{D}_s[\psi] = \sum \text{sources}\,(+) \;-\; \sum \text{sinks}\,(-)$$

All minimized together on random $(\vec{r}, E)$ points:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{s} w_s \left| \mathcal{D}_s[\psi](\vec{r}_i, E_i) \right|^2$$

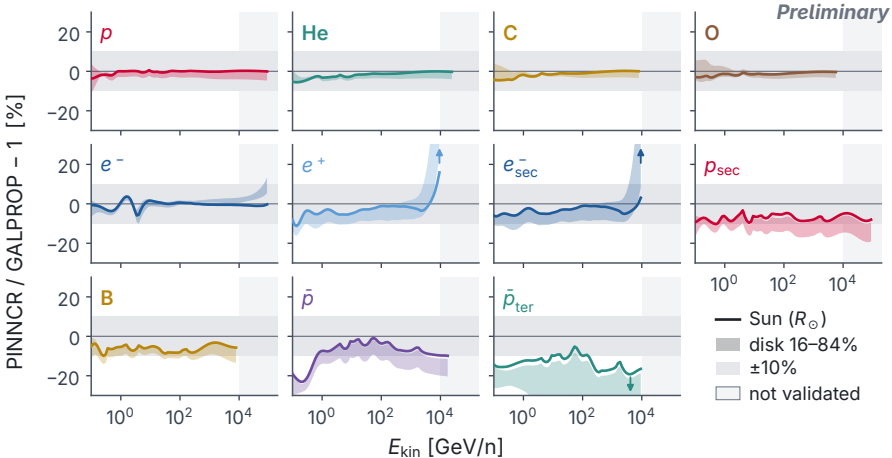
w_s: per-species weights that equalize the initial residuals.

Backup: Cost: the numbers

	GALPROP (CPU, 4 thr.)	PINN (1 GPU)
$e^- + p$, 0.1 kpc 3D grid†	$\sim$41 CPU-h, 1.5 TB	18 min train, 0.5 GB
$e^- + p$, largest measured (0.4 kpc)	$\sim$3 CPU-h, 95 GB	(same network)
11 sp. + frag., validation card (0.5 kpc)	19 CPU-h, 91 GB	2.2 h/seed, 5 GB
New point, same $\vec{\eta}$	interpolate stored grid	1.6–17 ns / 86 ns (11 sp.)
New $\vec{\eta}$	full re-solve	same forward pass
15-axis $\vec{\eta}$ cube, 11 sp.	16 CPU-h, 17 GB *per* $\vec{\eta}^\ddagger$	9 GPU-h/seed, 80 ns/pt
Stored artifact	$\sim$3.4 GB per $\vec{\eta}^\ddagger$	11 MB (10 seeds, whole cube)

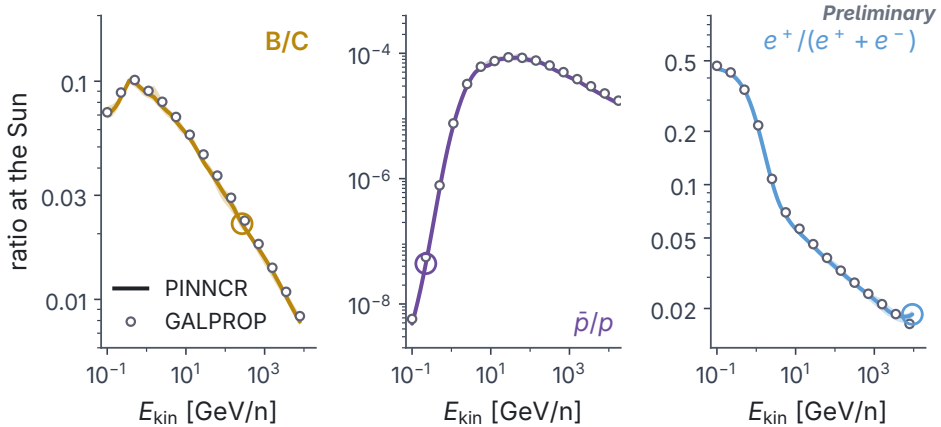
†extrapolated 16$\times$ beyond the largest run. ‡2D, 3 MeV floor, 17 species, 16 threads (CR solve only).

Backup: All 11 species vs. GALPROP



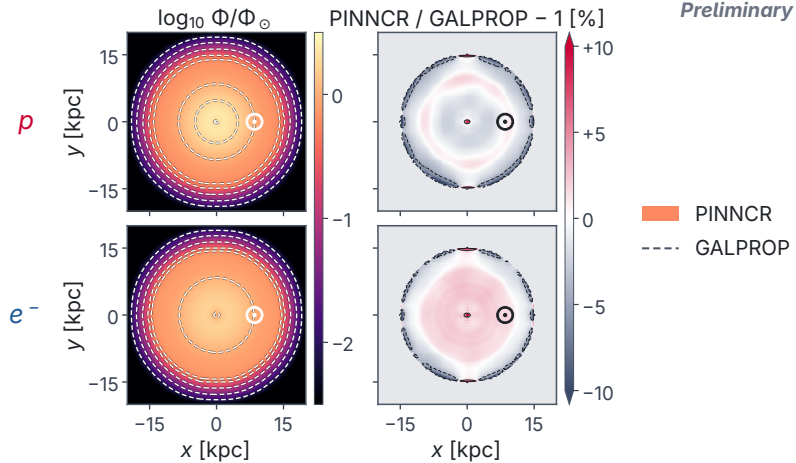
Primaries within 6%; secondaries 3–10% low, $\bar{p}$ worst below 1 GeV/n.
$e^{\pm}_{\rm sec}$ turn up only in the last bins before their clamp.

Backup: Secondary-to-primary ratios



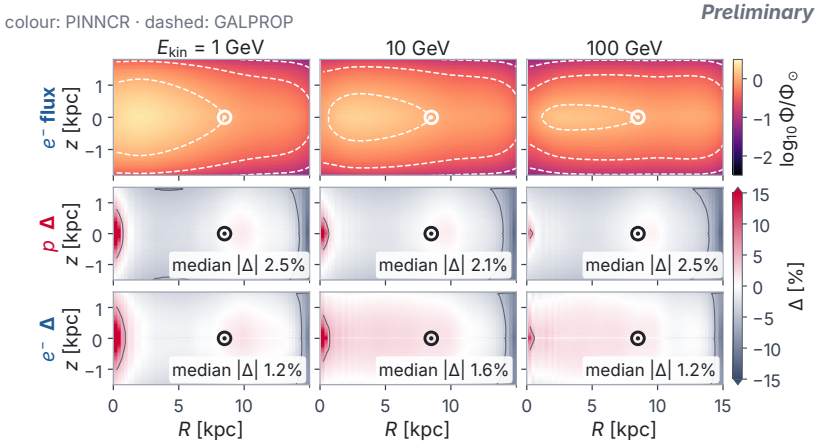
B/C median 5% (max 8%), $\bar{p}/p$ median 7% (max 22% at 0.2 GeV/n), e^+ fraction median 2%.
Median/max of |PINN/GALPROP-1| over the plotted range (ring = max); $\bar{p}/p$ uses secondary $\bar{p}$ only.

Backup: Morphology in the Galactic plane



$z = 0$, 10 GeV: median $|\Delta|$ 1.5% (p), 1.7% (e^-) inside $R < 15$ kpc; >5% only at the GC cusp ($R < 0.6$ kpc) and the outer rim.

Backup: Morphology in R–z



Median $|\Delta|$ 1.2–2.5% at 1, 10, 100 GeV, ∼90% of pixels within 5%. Breaks at the $R \to 0$ cusp (up to +50% at 1 GeV) and the $R = 15$ kpc edge (-17 to -20% along one axis; -6% azimuth-averaged).

Backup: GALPROP energy-grid convergence

		e^- bias at 1 TeV	
$f = E_{i+1}/E_i$	bins/decade	one-zone toy†	GALPROP (measured)
2.0	3.3	–	$+33\%$
1.4	6.8	$+48\%$	–
1.2 (default)	12.6	$+24\%$	$\approx$+9% (interp.)
1.1	24	$+12\%$	–
1.05	47	$+5.9\%$	$+2.5\%$
1.02	116	$+2.4\%$	–
1.01	231	$+1.2\%$	$+0.5\%$

†one-zone copy of GALPROP's scheme ($q \propto E^{-2.4}$). Measured = GALPROP vs. its $f \to 1$ limit; the main slide's +29% is vs. PINNCR.

- First-order loss stencil: error $\approx (\gamma_s/2) \ln f$ (γ_s = injection index) – mostly a normalization bias.
- Smears spectral features by $\sim$0.2 dex at $f = 1.2$; the top bin is a sink, keep $E_{\max}$ 1.5 dex above the analysis range.
- DRAGON2 (Evoli+17) and PICARD (Kissmann 14) see the same first- vs. second-order behaviour.

Backup: What the γ-ray timings cover

	PINN (10 seeds, 1 L40S)	GALPROP (2D, 12 thr.)
IC sky, 3°, 27 E_γ bins	22 s	93 min (82 min CR solve)
π^0+brems.+IC, 3°, renormalized	12.7 s	(same run)
15-axis sky Jacobian, 3° (primaries)	122 s	30 runs, $\sim$400 CPU-h
15-axis sky Jacobian, all species	$\sim$210 s	(same)
π^0+brems.+IC, nside 128, 16 bins	38 s / seed	–

- GALPROP's line-of-sight step alone is $\sim$11 min; the rest is the CR re-solve for every new $\vec{\eta}$.
- Fast IC path is exact for primaries; secondaries add $\sim$70%.
- Cost scales with pixels (2.3 ms/px): a 40°$\times$40° ROI at 0.5° is $\sim$15 s per map, $\sim$1 min per gradient (extrapolated).

Backup: What's in $\vec{\eta}$, and what isn't

Varied (15 axes)

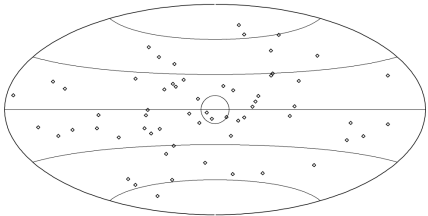
- Transport: D_0, δ, z_L, v_A, dv_c/dz.
- H and He injection: double-broken power laws, 5 parameters each; C, O follow H.
- Box: Cholis+21 models A–F, per-axis min/max $+8\%$.

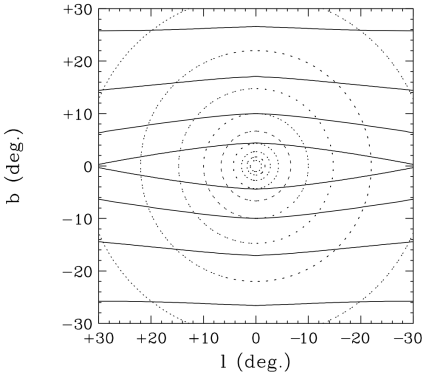
Fixed across the cube

- e^- injection, CR source distribution.
- Gas (GALPROP), ISRF (GALPROP standard), B-field.
- Uniform, isotropic, single power-law D (no breaks).
- Cross sections: GALPROP defaults.

- So today: marginalize over CR propagation, not the whole diffuse model.
- Next inputs to add: e^- injection, source distribution, ISRF level.

Backup: MSP morphology



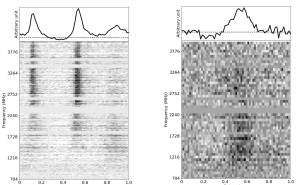


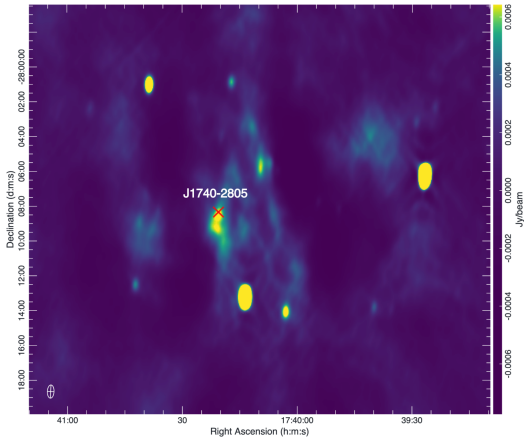
Hooper & Goodenough, arXiv:1407.5625

- A GC MSP population gives a different “boxy” morphology that traces the bulge + thick disk, rather than an annihilating DM halo.

Backup: Two new bulge MSPs

- Berteaud et al.: deep MeerKAT/GBT follow-up of inner-Galaxy X-ray sources.
- Two new MSPs: now 4 within 2° of the GC.
  - PSR J1740–2805
  - PSR J1740–28
- Matches predicted MSP detection rates in the bulge [Hooper, Sweden 2023].





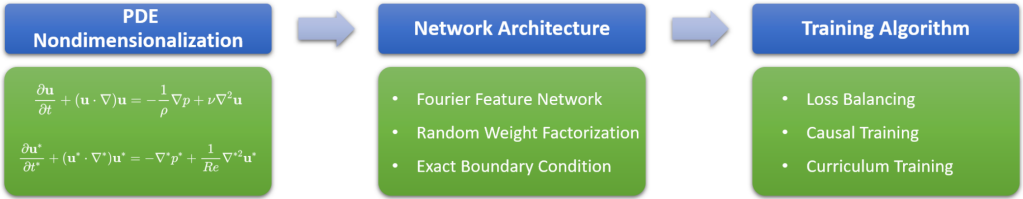
Berteaud et al., arXiv:2512.16699

Backup: Equilibrium for free

- GALPROP *time-evolves* initial conditions until $\partial_t \psi = 0$.
- Timescales run from $\gtrsim 10^8$ yr (low-E diffusion) to $\sim 3 \times 10^5$ yr (TeV $e^{\pm}$ losses): time steps down to $\lesssim 10^2$ yr, repeated hundreds of times [Kissmann+15].
- The PINN solves the steady-state equation *directly*.
- No timestep convergence to check.

$$\cancel{\frac{\partial \psi}{\partial t}} = q + D_{xx} \nabla^2 \psi - \frac{\partial}{\partial E}(\dot{E}\, \psi)$$

Backup: Training PINNs



Wang, Sankaran & Perdikaris, arXiv:2308.08468

- Non-dimensionalization of fields and PDE coefficients to $\mathcal{O}(1)$.
- Fourier features; random weight factorization; modified MLP / PirateNet.
- Loss balancing; curriculum learning; adaptive collocation-point sampling.

Backup: Log-ratio residual

- Terms span many decades: a plain MSE is dominated by the largest (e^- carried 99.97% of the loss).
- Split each residual into gains P_s (+) and sinks M_s (−), both scaled by the species' source at the Sun:

$$r_s = \log_{10} \frac{P_s + \varepsilon}{M_s + \varepsilon} \qquad (r_s = 0 \iff \text{gains balance sinks})$$

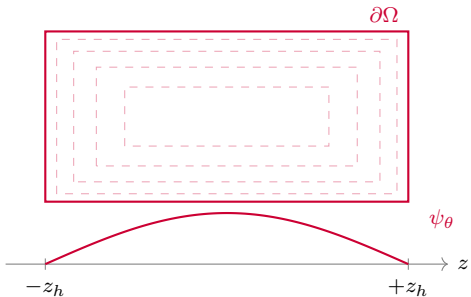
- $\mathcal{L} = \log_{10} \langle r_s^2 \rangle$ over points and species.
- Optional (hyperparameter choices):
  - weight active points, $r_s \to r_s \left[(P_s + M_s) + \varepsilon\right]^{\gamma/2}$;
  - per-species NTK trace balancing, $\lambda_s \propto 1/\mathrm{tr}\, K_s$.

Backup: Hard-coding the boundary condition

- CR halo: $\psi = 0$ on the boundary $\partial\Omega$ (free-escape).
- Factor the network as

$$\psi_\theta(\vec{r}, E) = B(\vec{r})\,\mathcal{N}_\theta(\vec{r}, E),$$

  B an approximate distance function (ADF): smooth, > 0 in Ω, $= 0$ on $\partial\Omega$.
- BC satisfied *by construction* – no boundary-penalty term to balance.



ADF level sets (dashed); ψ_θ vanishes at $\pm z_h$.

Backup: Factorized PINN architecture

Spectrum spans $\sim$8 decades; morphology varies slowly. Factor the solution:

$$\log_{10} \psi(\vec{r}, E) = \log_{10} \phi(\vec{r}, E) + \log_{10} S(E).$$

- $S(E)$ – global spectrum, energy only.
- $\phi(\vec{r}, E)$ – morphology, may carry residual E dependence.
- Both trained jointly $\Rightarrow$ percent-level agreement across all 8 decades.